Security Lab Solution Document

Infrastructure, Platform and Hardware Summary
Virtualization and Container Stack

HOMELAB
DEVSECOPS PLATFORM

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April 1, 2026

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Infrastructure, Platform and Hardware Summary

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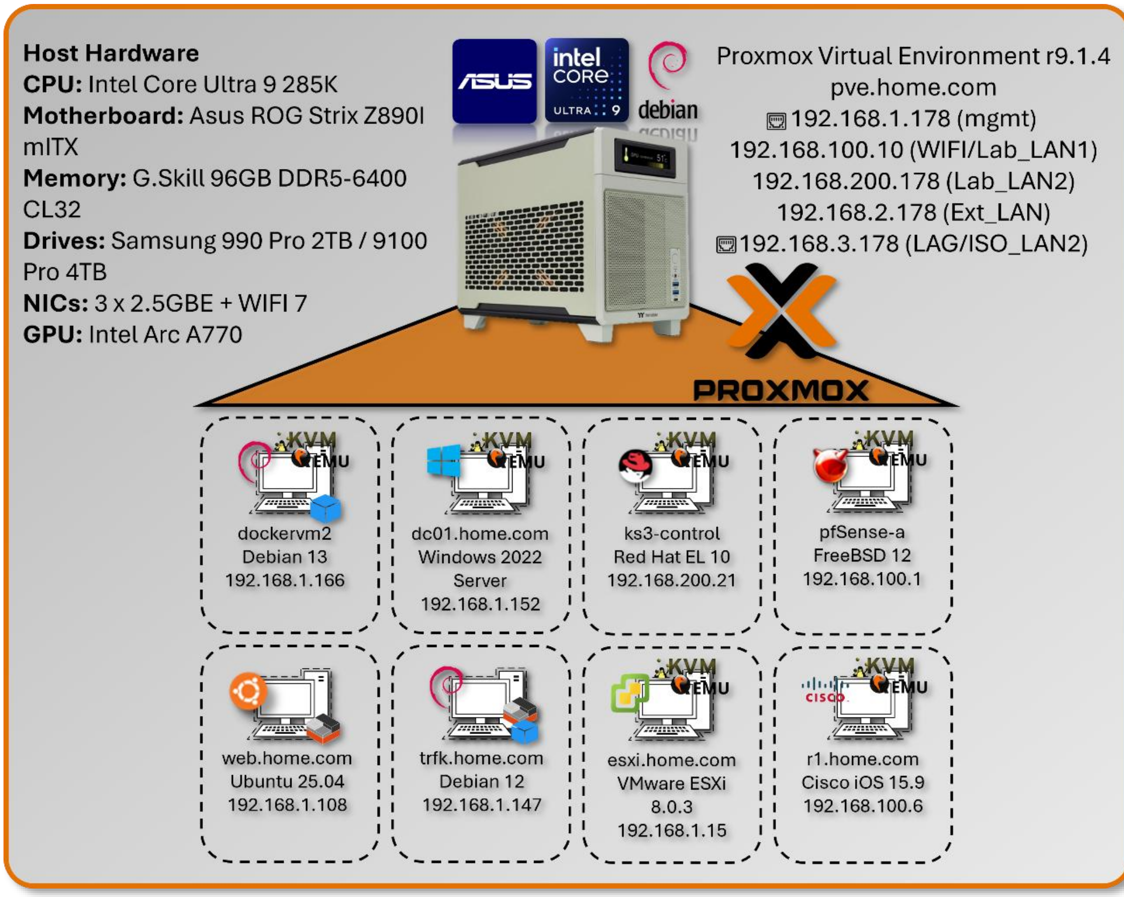


The lab's foundation is a single-node Proxmox Virtual Environment (VE) cluster running on enterprise-class bare-metal hardware, serving as the Type-1 hypervisor for the entire security operations platform. Proxmox provides KVM-based full virtualization for operating systems (Windows, BSD, Linux, MacOS) and LXC containerization for lightweight Linux workloads. The platform hosts approximately 30+ concurrent virtual machines and containers, supporting everything from high-availability pfSense firewalls to Kubernetes clusters, SIEM platforms, and malware analysis environments. Automated backup infrastructure via Proxmox Backup Server ensures rapid disaster recovery, while Proxmox Datacenter Manager provides centralized orchestration and monitoring.

Security Impact

- Bare-metal Type-1 hypervisor minimizes attack surface compared to Type-2 hosted solutions
- KVM hardware-assisted virtualization provides strong isolation between tenant workloads
- LXC containerization delivers near-native performance for network services with reduced overhead
- Snapshot and backup capabilities enable rapid rollback after security testing or compromise
- Centralized management reduces configuration drift and unauthorized system modifications
- ZFS storage backend provides data integrity verification and encryption at rest
- Network segmentation via virtual bridges isolates security zones (management, production, DMZ)

Deployment Rationale

Proxmox VE mirrors enterprise virtualization platforms (VMware vSphere, Microsoft Hyper-V, Red Hat Virtualization) while providing open-source flexibility and zero licensing costs. This enables building an enterprise-scale lab environment demonstrating production-grade infrastructure without commercial hypervisor expenses. Proxmox's integrated backup solution, clustering capabilities, and API-driven automation align with modern Infrastructure as Code (IaC) practices used in DevSecOps environments. The platform supports Docker and Kubernetes workloads alongside traditional VMs, demonstrating hybrid cloud architecture patterns increasingly common in enterprise security operations.

Architecture Principles Alignment

Defense in Depth:

- Hypervisor layer enforces hardware-based isolation between security zones
- Virtual network bridges segment traffic flows with firewall enforcement at each boundary
- Backup infrastructure resides on separate physical hosts (VMware Workstation, Synology NAS)
- Multiple authentication layers (Proxmox RBAC, Authentik SSO/OAuth2, VM-level access controls, application SSO)

Secure by Design:

- Default-deny network policies on all virtual bridges
- Mandatory TLS encryption for Proxmox web UI and API access
- Automated security updates via unattended-upgrades on Debian base OS
- PKI integration for certificate-based authentication across all services
- Immutable infrastructure through snapshot-based deployments

Zero Trust:

- No implicit trust between VMs or containers; all inter-service communication authenticated
- API access requires token-based authentication with scoped permissions
- Network policies enforce explicit allow rules; no "trusted" VLANs
- User access governed by role-based permissions (PVEAdmin, PVEAuditor, etc.)

Key Capabilities Demonstrated

Enterprise Infrastructure Operations:

- High-density virtualization supporting 30+ concurrent workloads on single node
- Mixed workload management (full VMs, containers, nested hypervisors and containers)
- Automated provisioning via Terraform and Ansible integration

Security Operations Platform:

- Isolated security tool deployment (SIEM, SOAR, threat intelligence, vulnerability scanners)
- Forensic analysis environments with snapshot-based evidence preservation
- Purple team infrastructure supporting both red and blue team operations

Disaster Recovery & Business Continuity:

- Automated weekly backups to Proxmox Backup Server with deduplication
- Off-host backup replication to Synology NAS for redundancy
- Rapid VM restoration (sub-15-minute RTO for critical services)
- Configuration-as-code enables full infrastructure rebuild from Git repository

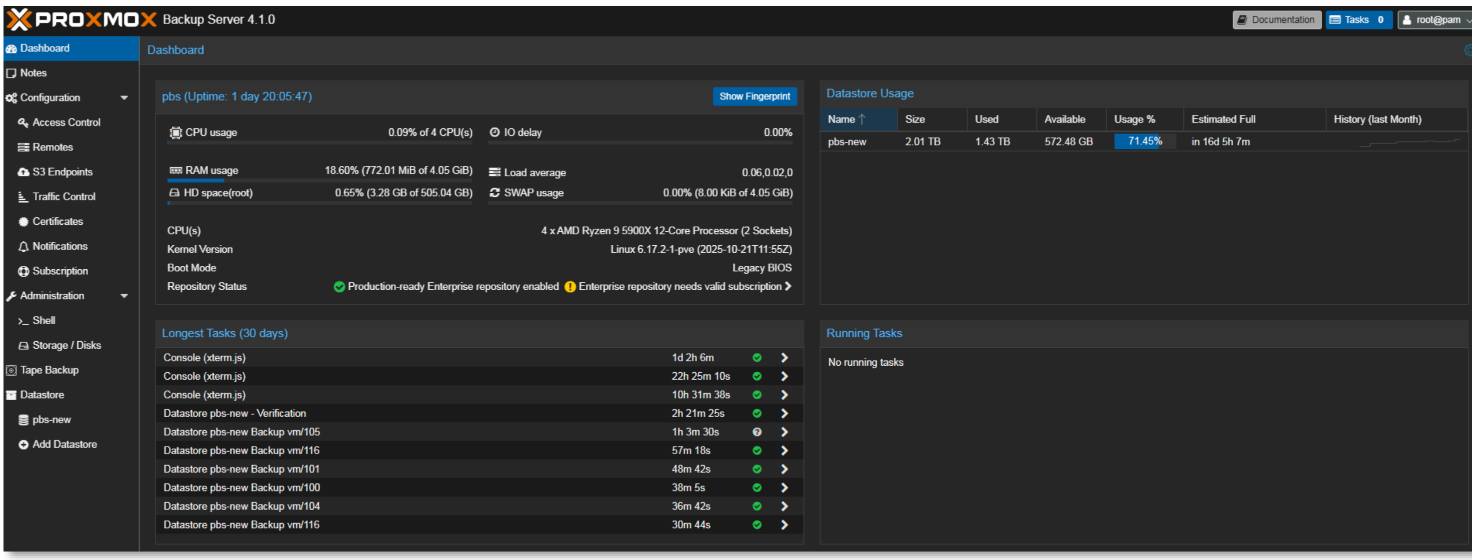
Advanced Storage & Networking:

- ZFS mirrored storage pools providing data integrity and high IOPS
- NVMe-based VM storage for database and container workloads
- Link Aggregation (LAG) with 802.3ad for high-throughput data transfers
- VLAN segmentation supporting isolated testing environments

Proxmox Backup Server

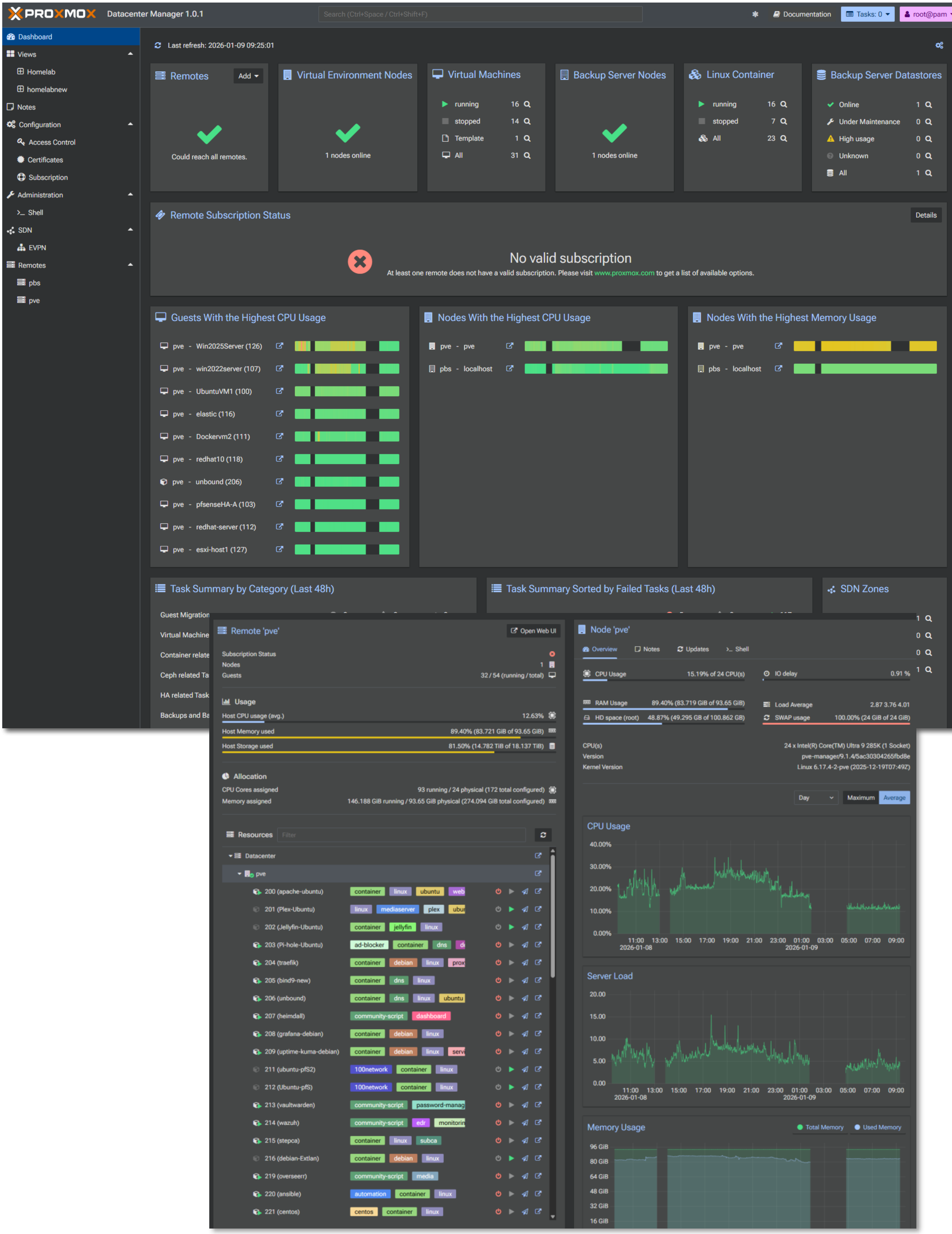
Integrated for automated, deduplicated backups of all virtual machines and containers. Backup jobs are scheduled to run weekly, with retention policies aligned to provide redundancies without taking up a lot of space since the server has been deployed within VMware Workstation running on my main production PC.

- Proxmox Backup Server deployed as nested VM within VMware Workstation
- Host: Production workstation (separate from lab infrastructure)
- Purpose: Off-host backup target with deduplication and compression
- Rationale: Leverages existing production hardware for cost-effective DR solution



Proxmox Datacenter Manager

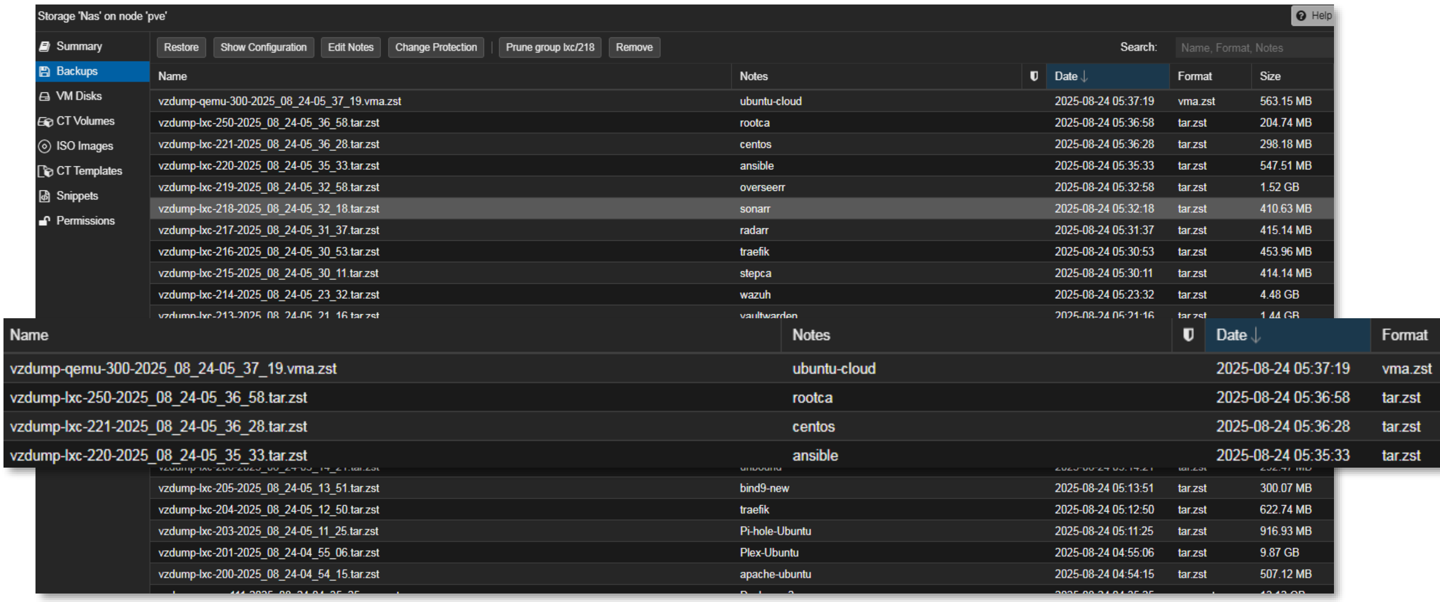
Centralized management solution to oversee and manage multiple nodes and clusters of Proxmox-based virtual environments


PROXMOX Datacenter Manager 1.0.1
Search (Ctrl+Space / Ctrl+Shift+F)
Documentation
Tasks: 0
root@pam
Dashboard
Views
Homelab
homelabnew
Notes
Configuration
Access Control
Certificates
Subscription
Administration
Shell
SDN
EVPN
Remotes
pbs
pve
Last refresh: 2026-01-09 09:25:01
Remotes
Add
Could reach all remotes.
Virtual Environment Nodes
1 nodes online
Virtual Machines
running 16
stopped 14
Template 1
All 31
Backup Server Nodes
1 nodes online
Linux Container
running 16
stopped 7
All 23
Backup Server Datastores
Online 1
Under Maintenance 0
High usage 0
Unknown 0
All 1
Remote Subscription Status
Details
No valid subscription
At least one remote does not have a valid subscription. Please visit www.proxmox.com to get a list of available options.
Guests With the Highest CPU Usage
pve - Win2025Server (126)
pve - win2022server (107)
pve - UbuntuVM1 (100)
pve - elastic (116)
pve - Dockervm2 (111)
pve - redhat10 (118)
pve - unbound (206)
pve - pfsenseHA-A (103)
pve - redhat-server (112)
pve - esxi-host1 (127)
Nodes With the Highest CPU Usage
pve - pve
pbs - localhost
Nodes With the Highest Memory Usage
pve - pve
pbs - localhost
Task Summary by Category (Last 48h)
Guest Migration
Virtual Machine
Container relate
Ceph related Ta
HA related Task
Backups and Ba
Task Summary Sorted by Failed Tasks (Last 48h)
SDN Zones
Remote 'pve'
Open Web UI
Subscription Status
Nodes 1
Guests 32 / 54 (running / total)
Usage
Host CPU usage (avg.) 12.63%
Host Memory used 89.40% (83.721 GiB of 93.65 GiB)
Host Storage used 81.50% (14.782 TiB of 18.137 TiB)
Allocation
CPU Cores assigned 93 running / 24 physical (172 total configured)
Memory assigned 146.188 GiB running / 93.65 GiB physical (274.094 GiB total configured)
Resources
Filter
Datacenter
pve
200 (apache-ubuntu)
201 (Plex-Ubuntu)
202 (Jellyfin-Ubuntu)
203 (Pi-hole-Ubuntu)
204 (traefik)
205 (bind9-new)
206 (unbound)
207 (heimdall)
208 (grafana-debian)
209 (uptime-kuma-debian)
211 (ubuntu-pfS2)
212 (Ubuntu-pfS)
213 (vaultwarden)
214 (wazuh)
215 (stepca)
216 (debian-Extlan)
219 (overseerr)
220 (ansible)
221 (centos)
Node 'pve'
Overview
Notes
Updates
Shell
CPU Usage 15.19% of 24 CPU(s)
IO delay 0.91 %
RAM Usage 89.40% (83.719 GiB of 93.65 GiB)
Load Average 2.87 3.76 4.01
HD space (root) 48.87% (49.295 GiB of 100.862 GiB)
SWAP usage 100.00% (24 GiB of 24 GiB)
CPU(s) 24 x Intel(R) Core(TM) Ultra 9 285K (1 Socket)
Version pve-manager/9.1.4/5ac30304265fbd8e
Kernel Version Linux 6.17.4-2-pve (2025-12-19T07:49Z)
Day
Maximum
Average
CPU Usage
Server Load
Memory Usage
Total Memory
Used Memory

Physical Network Attached Storage (NAS) Integration:

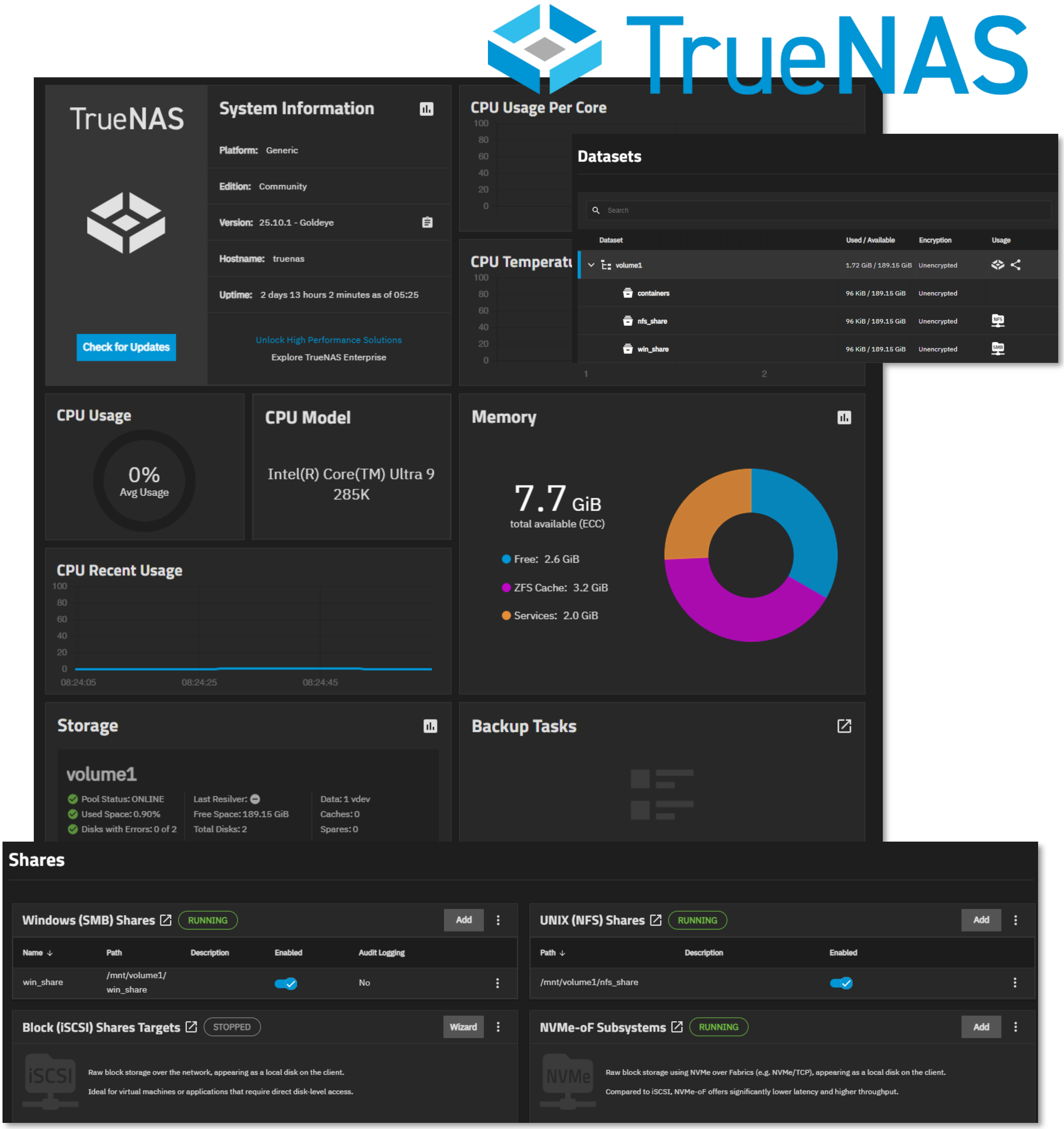
A shared Synology NAS is configured to receive automated backups from the Proxmox Backup Server. This ensures off-host redundancy and supports rapid restoration in case of lab-wide failure or rollback testing.

- Target: Synology NAS (DSM 6.x) via SMB/NFS mount
- Encryption: AES-256 at rest, TLS in transit
- Purpose: rapid VM restoration, long-term archival, K3s off-host PVCs



Virtual Network Attached Storage (NAS) Integration:

Promox virtual machine running TrueNAS to support Windows (SMB) and Linux (NFS) mounts for data sharing and redundancy. The single storage pool is configured for mirroring across the two NVMe drives.



Proxmox Host Hardware

Component	Specification	Justification
CPU	Intel Core Ultra 9 285K (24c)	Performance/efficiency cores for mixed workloads
Cooling	AIO liquid cooling	Sustained thermal management under load
Memory	G.Skill 96 GB DDR5-6400 CL32	High-density RAM for ~50 concurrent VMs/containers
OS Storage	Samsung 990 PRO 2TB (PCIe 4.0)	Low-latency boot and snapshot operations
VM Storage	Samsung 9100 PRO 4TB (PCIe 5.0)	High IOPS for database and container workloads
Network Interfaces	3× 2.5GbE + WiFi 7	Redundant connectivity and traffic segregation
GPU	Intel Arc A770 (16GB)	PCIe passthrough for transcoding
Motherboard	ASUS ROG Strix Z890-I (mITX)	Compact form factor with enterprise features

Network Interface Design

Physical Interfaces:

- eth0 (2.5GbE): Proxmox management network (192.168.1.x)
- eth1 (2.5GbE): LAG member - VLAN3 trunk (192.168.3.x)
- eth2 (2.5GbE): LAG member - VLAN3 trunk (192.168.3.x)
- wlan0 (WiFi 7): Bridged to Lab_LAN1 virtual network

Virtual Bridges:

- vmbr0: Management and Prod_LAN workloads (192.168.1.x) (default gateway)
- vmbr1: Lab_LAN1 network workloads (192.168.100.x)
- vmbr2: LAG bond0 ISO_LAN2 network workloads (192.168.3.x)
- vmbr3: Lab_LAN2 network workloads (192.168.200.x)
- vmbr4: Ext_LAN2 network workloads (192.168.2.x)
- vmbr5: Ext_LAN network workloads (10.20.0.x)
- vmbr7: pfSense HA Sync (10.10.0.x)
- vmbr8: Cisco Point-to-Point Network (10.30.0.x)

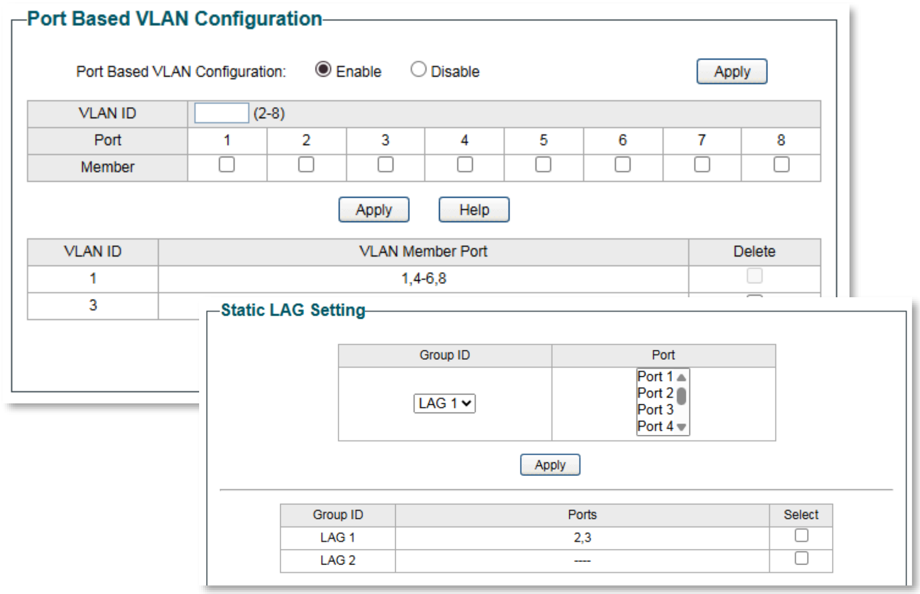
Lab Switch

TP-Link TL-SG108E Smart Switch

Used for VLAN and Link Aggregation (LAG) testing between The Proxmox host server, FortiGate firewall and a Windows 11 Pro workstation.

Configuration:

- Link Aggregation: Static LAG (802.3ad equivalent) on ports 2-3
- Lab server NICs 1-2 bonded (LACP mode balance-rr)
- Aggregate bandwidth: 2 Gbps



VLAN Segmentation:

- VLAN 3 (192.168.3.0/24): Isolated testing subnet
- Ports 2, 3, 7: Tagged VLAN3 members
- Purpose: Dedicated high-speed link between office PC and lab server
- Use case: Large file transfers, iSCSI testing, backup replication

Proxmox Workload Overview

The majority of hosts and services run within the Proxmox environment and run within one of the two integrated technologies supported:

Virtual Machines (VMs)

- **Tech Stack**: KVM (Kernel-based Virtual Machine) + QEMU
- **Use Case**: Full OS virtualization—ideal for Windows, BSD, or isolated Linux environments

Linux Containers (LXC)

- **Tech Stack**: LXC (Linux Containers)
- **Use Case**: Lightweight virtualization for Linux apps with near-native performance

Table A -Infrastructure Asset Summary Table

Table B – Application and Services Summary Table

Infrastructure Visualization

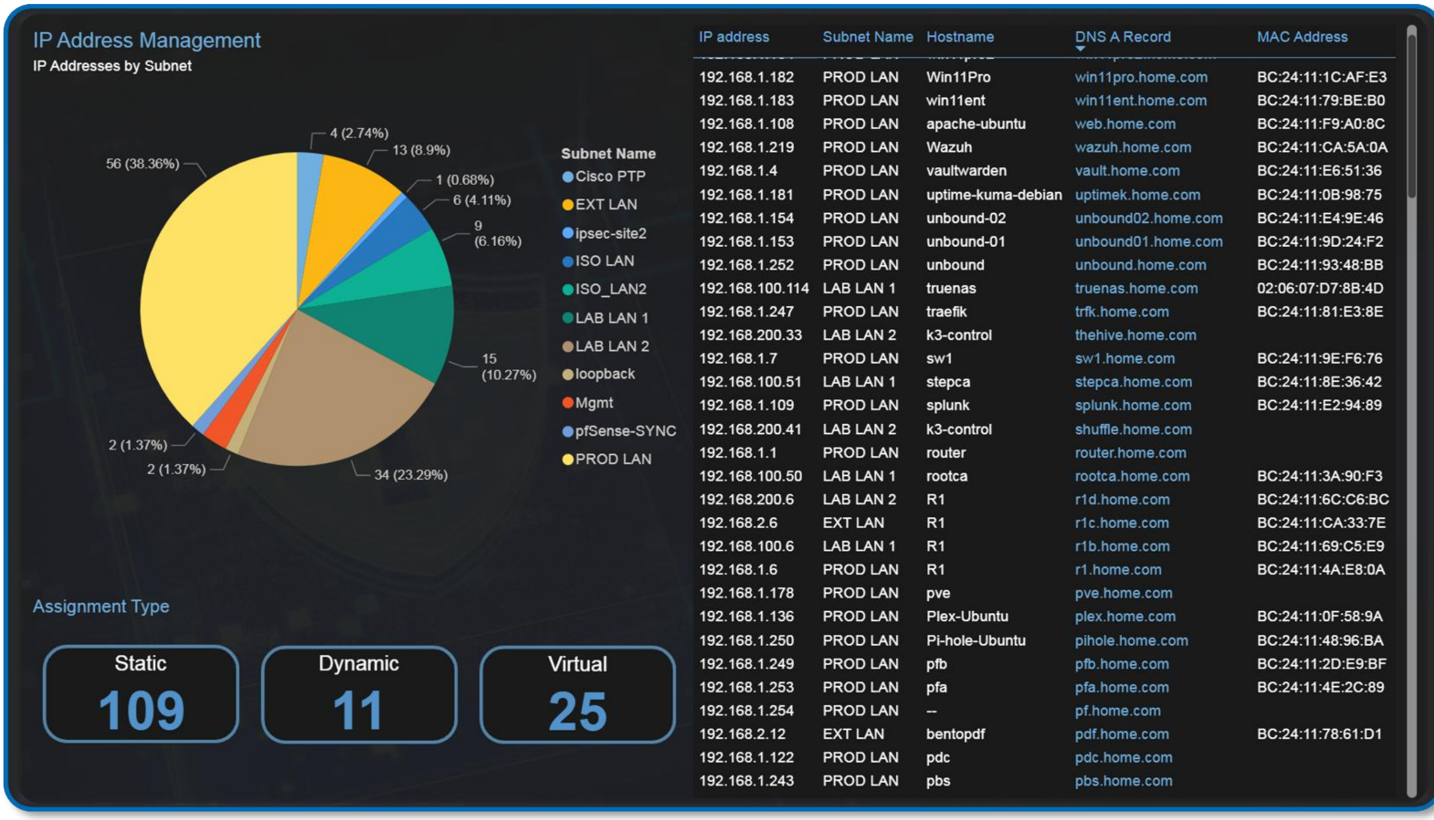
PowerBI Dashboards

Custom-built Power BI reports provide real-time visibility into lab operations:

Application Inventory: Per-host service mapping with version tracking

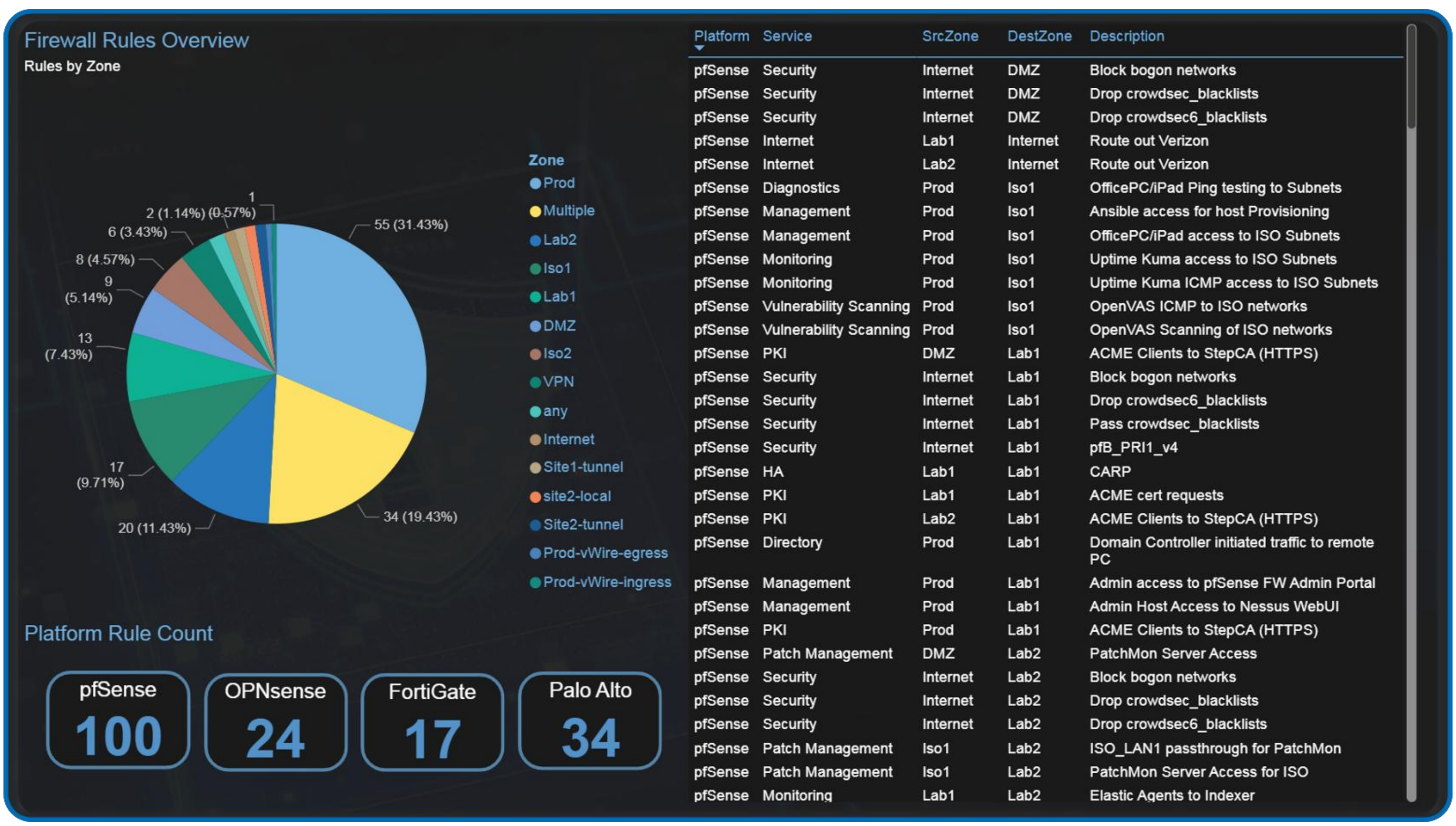
- Resource Utilization: CPU/memory/storage trends across Proxmox nodes
- IP Address Management: Assigned IP addresses by subnet and assignment type
- OS Distribution Analysis: Platform breakdown (RHEL, Ubuntu, Debian, Windows)

IP Address Overview



IP address	Subnet Name	Hostname	DNS A Record	MAC Address
192.168.1.182	PROD LAN	Win11Pro	win11pro.home.com	BC:24:11:1C:AF:E3
192.168.1.183	PROD LAN	win11ent	win11ent.home.com	BC:24:11:79:BE:B0
192.168.1.108	PROD LAN	apache-ubuntu	web.home.com	BC:24:11:F9:A0:8C
192.168.1.219	PROD LAN	Wazuh	wazuh.home.com	BC:24:11:CA:5A:0A
192.168.1.4	PROD LAN	vaultwarden	vault.home.com	BC:24:11:E6:51:36
192.168.1.181	PROD LAN	uptime-kuma-debian	uptimek.home.com	BC:24:11:0B:98:75
192.168.1.154	PROD LAN	unbound-02	unbound02.home.com	BC:24:11:E4:9E:46
192.168.1.153	PROD LAN	unbound-01	unbound01.home.com	BC:24:11:9D:24:F2
192.168.1.252	PROD LAN	unbound	unbound.home.com	BC:24:11:93:48:BB
192.168.100.114	LAB LAN 1	truenas	truenas.home.com	02:06:07:D7:8B:4D
192.168.1.247	PROD LAN	traefik	trfk.home.com	BC:24:11:81:E3:8E
192.168.200.33	LAB LAN 2	k3-control	thehive.home.com	
192.168.1.7	PROD LAN	sw1	sw1.home.com	BC:24:11:9E:F6:76
192.168.100.51	LAB LAN 1	stepca	stepca.home.com	BC:24:11:8E:36:42
192.168.1.109	PROD LAN	splunk	splunk.home.com	BC:24:11:E2:94:89
192.168.200.41	LAB LAN 2	k3-control	shuffle.home.com	
192.168.1.1	PROD LAN	router	router.home.com	
192.168.100.50	LAB LAN 1	rootca	rootca.home.com	BC:24:11:3A:90:F3
192.168.200.6	LAB LAN 2	R1	r1d.home.com	BC:24:11:6C:C6:BC
192.168.2.6	EXT LAN	R1	r1c.home.com	BC:24:11:CA:33:7E
192.168.100.6	LAB LAN 1	R1	r1b.home.com	BC:24:11:69:C5:E9
192.168.1.6	PROD LAN	R1	r1.home.com	BC:24:11:4A:E8:0A
192.168.1.178	PROD LAN	pve	pve.home.com	
192.168.1.136	PROD LAN	Plex-Ubuntu	plex.home.com	BC:24:11:0F:58:9A
192.168.1.250	PROD LAN	Pi-hole-Ubuntu	pihole.home.com	BC:24:11:48:96:BA
192.168.1.249	PROD LAN	pfb	pfb.home.com	BC:24:11:2D:E9:BF
192.168.1.253	PROD LAN	pfa	pfa.home.com	BC:24:11:4E:2C:89
192.168.1.254	PROD LAN	--	pf.home.com	
192.168.2.12	EXT LAN	bentopdf	pdf.home.com	BC:24:11:78:61:D1
192.168.1.122	PROD LAN	pdc	pdc.home.com	
192.168.1.243	PROD LAN	pbs	pbs.home.com	

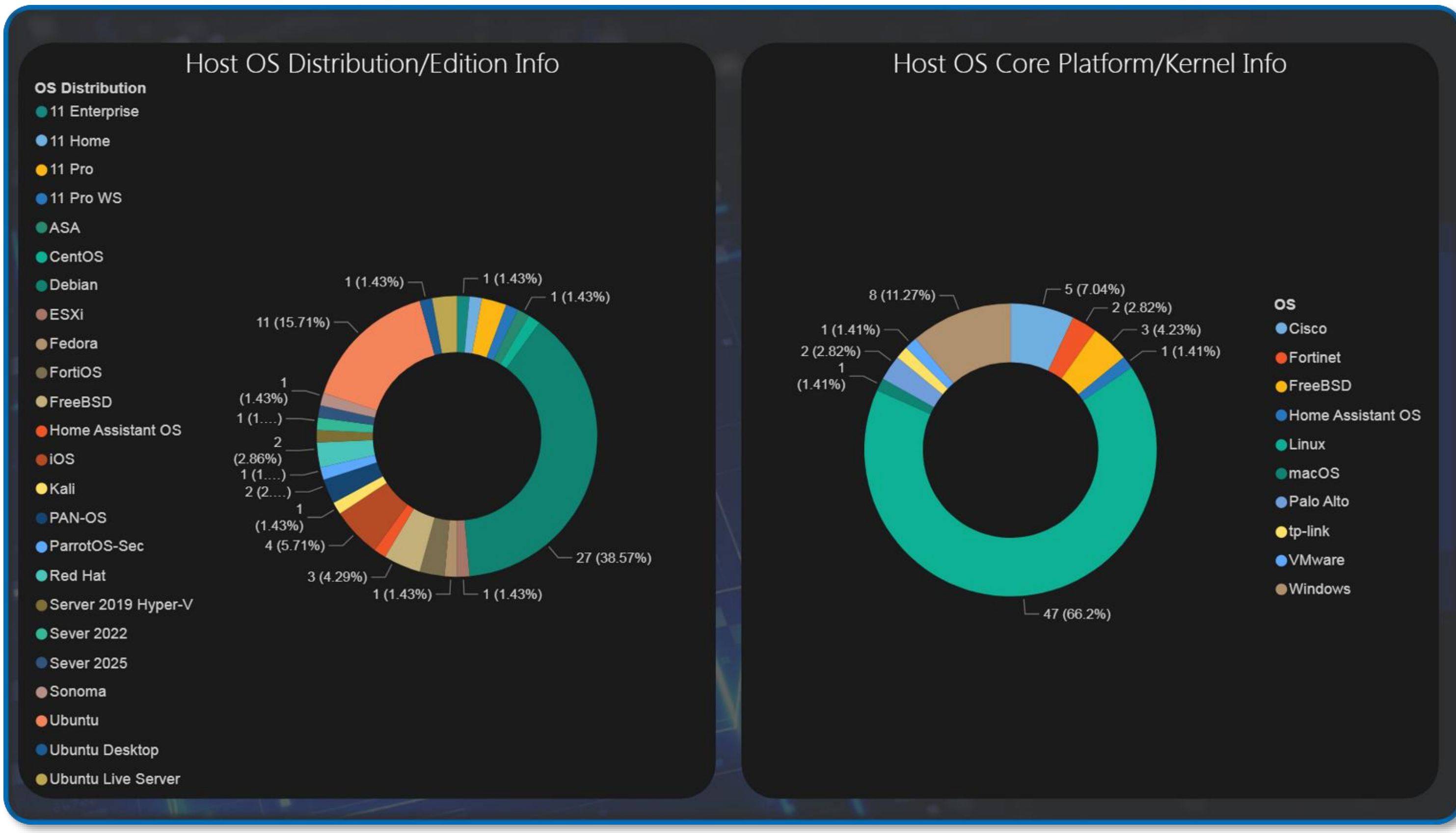
Firewall Rule Overview



Platform	Service	SrcZone	DestZone	Description
pfSense	Security	Internet	DMZ	Block bogon networks
pfSense	Security	Internet	DMZ	Drop crowdsec_blacklists
pfSense	Security	Internet	DMZ	Drop crowdsec6_blacklists
pfSense	Internet	Lab1	Internet	Route out Verizon
pfSense	Internet	Lab2	Internet	Route out Verizon
pfSense	Diagnostics	Prod	Iso1	OfficePC/iPad Ping testing to Subnets
pfSense	Management	Prod	Iso1	Ansible access for host Provisioning
pfSense	Management	Prod	Iso1	OfficePC/iPad access to ISO Subnets
pfSense	Monitoring	Prod	Iso1	Uptime Kuma access to ISO Subnets
pfSense	Monitoring	Prod	Iso1	Uptime Kuma ICMP access to ISO Subnets
pfSense	Vulnerability Scanning	Prod	Iso1	OpenVAS ICMP to ISO networks
pfSense	Vulnerability Scanning	Prod	Iso1	OpenVAS Scanning of ISO networks
pfSense	PKI	DMZ	Lab1	ACME Clients to StepCA (HTTPS)
pfSense	Security	Internet	Lab1	Block bogon networks
pfSense	Security	Internet	Lab1	Drop crowdsec6_blacklists
pfSense	Security	Internet	Lab1	Pass crowdsec_blacklists
pfSense	Security	Internet	Lab1	pfB_PRI1_v4
pfSense	HA	Lab1	Lab1	CARP
pfSense	PKI	Lab1	Lab1	ACME cert requests
pfSense	PKI	Lab2	Lab1	ACME Clients to StepCA (HTTPS)
pfSense	Directory	Prod	Lab1	Domain Controller initiated traffic to remote PC
pfSense	Management	Prod	Lab1	Admin access to pfSense FW Admin Portal
pfSense	Management	Prod	Lab1	Admin Host Access to Nessus WebUI
pfSense	PKI	Prod	Lab1	ACME Clients to StepCA (HTTPS)
pfSense	Patch Management	DMZ	Lab2	PatchMon Server Access
pfSense	Security	Internet	Lab2	Block bogon networks
pfSense	Security	Internet	Lab2	Drop crowdsec_blacklists
pfSense	Security	Internet	Lab2	Drop crowdsec6_blacklists
pfSense	Patch Management	Iso1	Lab2	ISO_LAN1 passthrough for PatchMon
pfSense	Patch Management	Iso1	Lab2	PatchMon Server Access for ISO
pfSense	Monitoring	Lab1	Lab2	Elastic Agents to Indexer

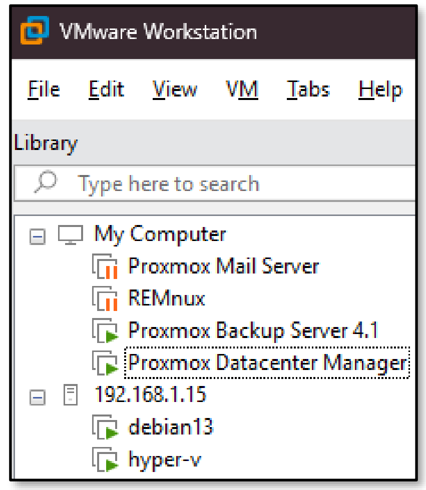
[Proxmox Node PVE Summary]()

OS Platform and Distribution/Edition Summary



VMware vSphere r8 Environment

Deployment Overview



The lab operates a hybrid virtualization architecture combining Proxmox as the primary Type-1 hypervisor with VMware vSphere r8 for specialized workloads. VMware Workstation Pro 25H2 runs on the production workstation, hosting critical backup infrastructure (Proxmox Backup Server, Datacenter Manager, Mail Gateway) and malware analysis environments (REMnux). ESXi r8 hypervisor provides enterprise-grade virtualization for Windows Server 2019 Hyper-V nested environments and Debian Live systems.

Security Impact

- Isolated backup infrastructure prevents contamination of production Proxmox environment
- REMnux sandbox provides safe malware analysis without risking lab infrastructure
- Hyper-V nested virtualization enables Windows-specific security testing (AD exploitation, PowerShell analysis)
- Multi-hypervisor architecture reduces vendor lock-in risk and single-point-of-failure scenarios

Deployment Rationale

Enterprise environments frequently operate multiple virtualization platforms where VMware handles legacy workloads, specific compliance requirements, or vendor-mandated infrastructure. This deployment demonstrates proficiency with multi-vendor hypervisor management, cross-platform migration strategies, and vendor-neutral infrastructure design. Running Proxmox Backup Server on VMware Workstation mirrors enterprise DR architectures where backup infrastructure resides on separate physical hosts or geographic locations.

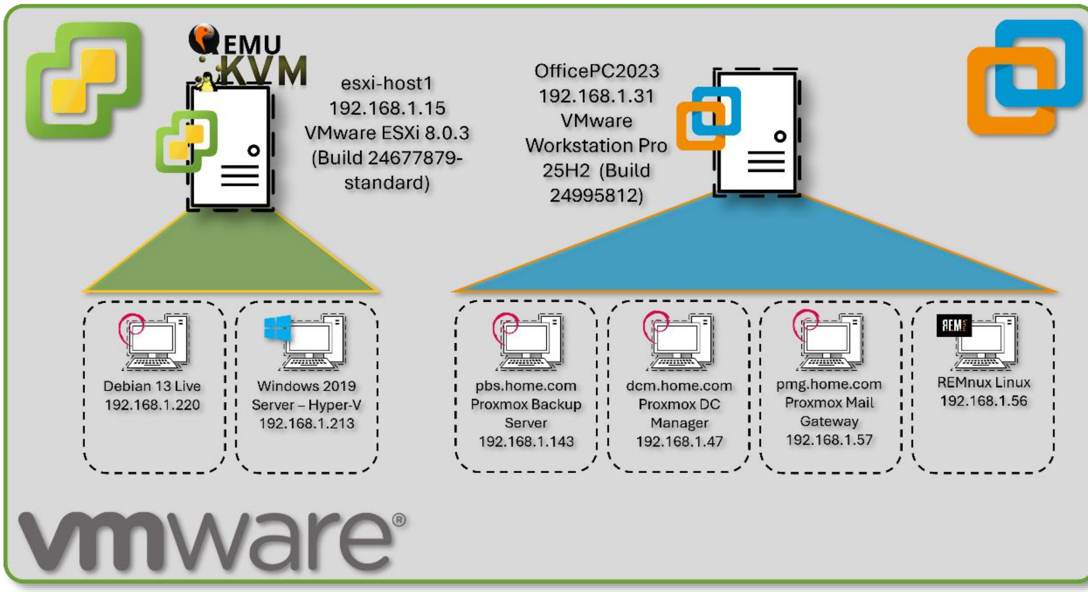
Architecture Principles Alignment

- **Defense in Depth:** Backup infrastructure physically separated from production hypervisor; malware analysis isolated in disposable VMs
- **Secure by Design:** Nested virtualization enforces additional isolation layers; backup encryption at rest and in transit
- **Zero Trust:** No implicit trust between hypervisor platforms; each VM authenticated independently

Configuration Details

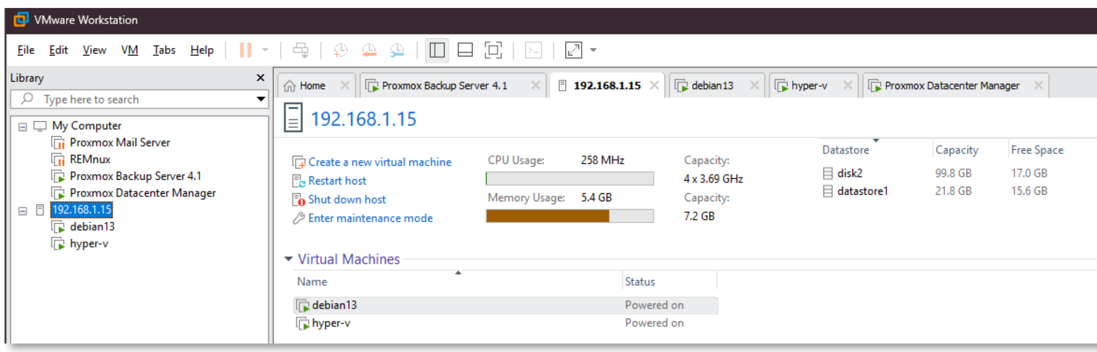
VMware Workstation Pro 25H2 (Build 24995812):

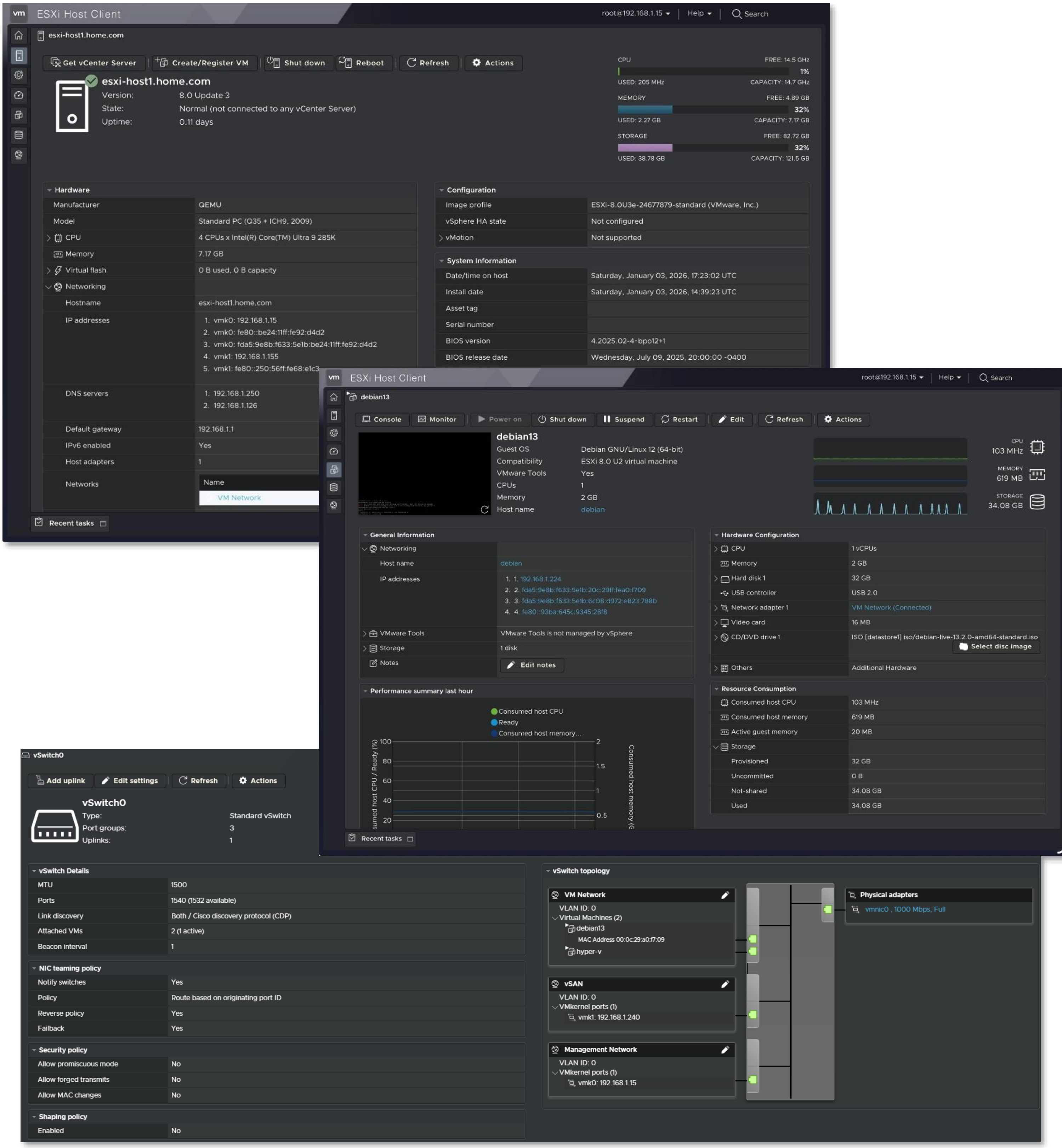
- Proxmox Backup Server v4.1.0 - Deduplicated backup target with AES-256 encryption
- Proxmox Datacenter Manager v1.0.1 - Centralized multi-cluster orchestration
- Proxmox Mail Gateway - Email security gateway testing
- REMnux Linux - Malware analysis and reverse engineering toolkit



ESXi r8 (Build 24677879-standard):

- Windows Server 2019 Hyper-V - Nested hypervisor for AD security research
- Debian Live r13 - Ephemeral forensics and incident response platform




ESXi Host Client
root@192.168.1.15
Help
Search
esxi-host1.home.com
Get vCenter Server
Create/Register VM
Shut down
Reboot
Refresh
Actions
esxi-host1.home.com
Version: 8.0 Update 3
State: Normal (not connected to any vCenter Server)
Uptime: 0.11 days
CPU FREE: 14.5 GHz 1% USED: 205 MHz CAPACITY: 14.7 GHz
MEMORY FREE: 4.89 GB 32% USED: 2.27 GB CAPACITY: 7.17 GB
STORAGE FREE: 82.72 GB 32% USED: 38.78 GB CAPACITY: 121.5 GB
Hardware
Manufacturer QEMU
Model Standard PC (Q35 + ICH9, 2009)
CPU 4 CPUs x Intel(R) Core(TM) Ultra 9 285K
Memory 7.17 GB
Virtual flash 0 B used, 0 B capacity
Networking
Hostname esxi-host1.home.com
IP addresses
1. vmk0: 192.168.1.15
2. vmk0: fe80::be24:11ff:fe92:d4d2
3. vmk0: fda5:9e8b:f633:5e1b:be24:11ff:fe92:d4d2
4. vmk1: 192.168.1.155
DNS servers
1. 192.168.1.250
2. 192.168.1.126
Default gateway 192.168.1.1
IPv6 enabled Yes
Host adapters 1
Networks
Name
VM Network
Recent tasks
Configuration
Image profile ESXi-8.0U3e-24677879-standard (VMware, Inc.)
vSphere HA state Not configured
vMotion Not supported
System Information
Date/time on host Saturday, January 03, 2026, 17:23:02 UTC
Install date Saturday, January 03, 2026, 14:39:23 UTC
Asset tag
Serial number
BIOS version 4.2025.02-4-bpo12+1
BIOS release date Wednesday, July 09, 2025, 20:00:00 -0400
ESXi Host Client
debian13
Console
Monitor
Power on
Shut down
Suspend
Restart
Edit
Refresh
Actions
debian13
Guest OS Debian GNU/Linux 12 (64-bit)
Compatibility ESXi 8.0 U2 virtual machine
VMware Tools Yes
CPUs 1
Memory 2 GB
Host name debian
CPU 103 MHz
MEMORY 619 MB
STORAGE 34.08 GB
General Information
Networking
Host name debian
IP addresses
1. 192.168.1.224
2. fda5:9e8b:f633:5e1b:20c:29ff:fea0:f709
3. fda5:9e8b:f633:5e1b:6c08:d972:e823:788b
4. fe80::93ba:645c:9345:28f8
VMware Tools VMware Tools is not managed by vSphere
Storage 1 disk
Notes
Edit notes
Hardware Configuration
CPU 1 vCPUs
Memory 2 GB
Hard disk 1 32 GB
USB controller USB 2.0
Network adapter 1 VM Network (Connected)
Video card 16 MB
CD/DVD drive 1 ISO [datastore1] iso/debian-live-13.2.0-amd64-standard.iso
Select disc image
Others Additional Hardware
Performance summary last hour
Consumed host CPU
Ready
Consumed host memory...
Resource Consumption
Consumed host CPU 103 MHz
Consumed host memory 619 MB
Active guest memory 20 MB
Storage
Provisioned 32 GB
Uncommitted 0 B
Not-shared 34.08 GB
Used 34.08 GB
Recent tasks
vSwitch0
Add uplink
Edit settings
Refresh
Actions
vSwitch0
Type: Standard vSwitch
Port groups: 3
Uplinks: 1
vSwitch Details
MTU 1500
Ports 1540 (1532 available)
Link discovery Both / Cisco discovery protocol (CDP)
Attached VMs 2 (1 active)
Beacon interval 1
NIC teaming policy
Notify switches Yes
Policy Route based on originating port ID
Reverse policy Yes
Failback Yes
Security policy
Allow promiscuous mode No
Allow forged transmits No
Allow MAC changes No
Shaping policy
Enabled No
vSwitch topology
VM Network
VLAN ID: 0
Virtual Machines (2)
debian13
MAC Address 00:0c:29:a0:f7:09
hyper-v
vSAN
VLAN ID: 0
VMkernel ports (1)
vmk1: 192.168.1.240
Management Network
VLAN ID: 0
VMkernel ports (1)
vmk0: 192.168.1.15
Physical adapters
vmnic0 , 1000 Mbps, Full

Cisco Physical and Virtual Infrastructure

Deployment Overview

The lab operates a mixed Cisco infrastructure consisting of three virtual IOS XE/IOS routers running as KVM guests in Proxmox and one physical Catalyst 3560X Layer 3 switch. R1 and R2 are vIOS instances (IOS 15.9(3)M6), R3 is a CSR1000V running IOS XE 16.12.5, and the WS-C3560X-24T-S runs IOS 15.0(2)SE. All devices participate in OSPF Area 0 for dynamic route distribution.

Security Impact

- Isolated routing domain — virtual and physical Cisco infrastructure creates a contained routing environment; protocol testing, ACL validation, and failure scenario simulation do not impact production networks
- Policy sandbox — tests ACLs, prefix lists, route maps, and firewall rules before deployment to pfSense/OPNsense/FortiGate
- Attack path analysis — multi-hop topology enables lateral movement detection and network segmentation validation
- Layer 2 hardening — DHCP snooping, BPDU guard, port-security, and sticky MACs on the physical switch prevent common L2 attacks (ARP spoofing, rogue DHCP, MAC flooding)
- Telemetry coverage — NetFlow export from three devices to ntopng provides flow-level visibility across all major lab segments
- OSPF MD5 authentication — prevents route injection from unauthorized neighbors

Deployment Rationale

Cisco IOS and IOS XE power the majority of enterprise routers and Layer 3 switches globally. The mixed virtual/physical deployment demonstrates CLI proficiency, dynamic routing protocol implementation, and network security hardening across multiple IOS generations. The physical 3560X replaces the decommissioned vSwitch, adding real L2 port-density, hardware-based spanning tree, and physical port-security capabilities that virtual switches cannot fully replicate. Running vIOS (R1, R2) and a CSR1000V (R3) within Proxmox KVM enables integration with Ansible network modules, Netmiko, and NAPALM for network automation workflows.

Architecture Principles Alignment

Defense in Depth

- Virtual routing domain physically isolated from production routing
- OSPF MD5 authentication on all peering interfaces
- ACLs enforced at VTY lines; management restricted to Prod_LAN subnet
- Physical switch DHCP snooping and BPDU guard enforce L2 boundary integrity

Secure by Design

- SSH v2 only across all four devices; cipher and MAC algorithms explicitly hardened
- Type 9 (SCRYPT) secrets on IOS XE devices; service password-encryption on all
- HTTP management disabled; no passive management plane exposure
- Unused switch ports shut down and assigned to a dead VLAN

Zero Trust

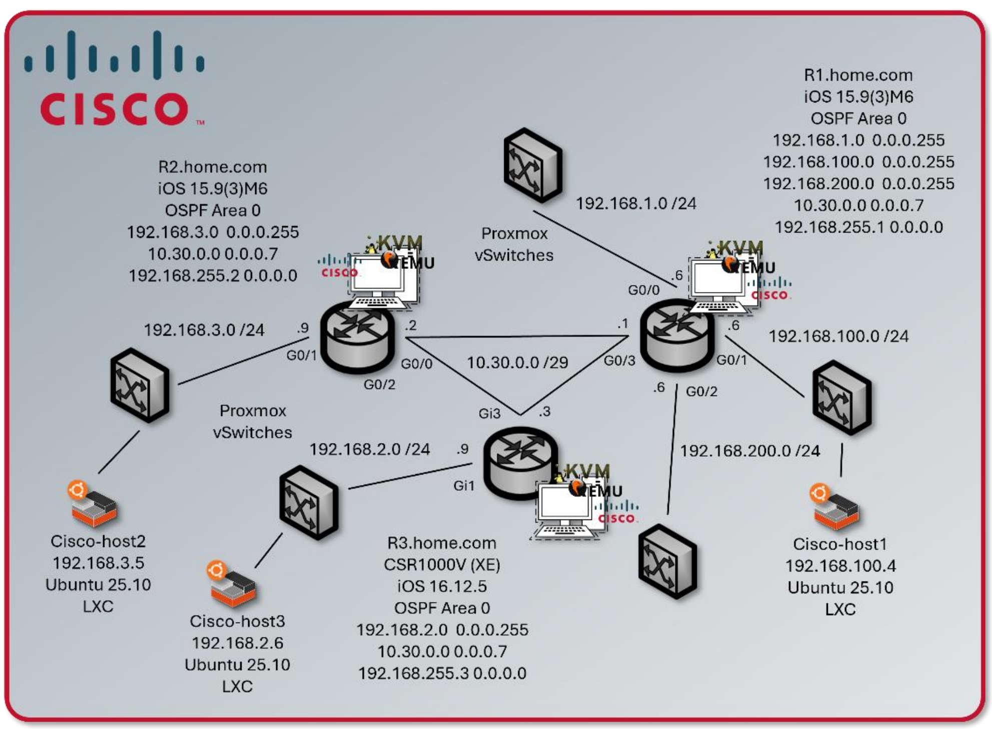
- No implicit routing trust; all OSPF neighbors explicitly configured with authentication

- Each device authenticates independently to management infrastructure via local user database
- Inter-VLAN routing on the 3560X requires explicit SVI configuration and static routing — no implicit transit

Device Inventory

Hostname	Model / Platform	IOS Version	Role	Management IP
r1	Cisco vIOS (KVM/Proxmox)	IOS 15.9(3)M6	Primary router — gateway & NetFlow export	192.168.1.6
r2	Cisco vIOS (KVM/Proxmox)	IOS 15.9(3)M6	Secondary router — isolated test network & NetFlow export	192.168.3.9
r3	CSR1000V (KVM/Proxmox)	IOS XE 16.12.5	Edge router — DMZ/Ext & NetFlow export	192.168.2.9
cisco-3560	WS-C3560X-24T-S (Physical)	IOS 15.0(2)SE	Core L2/L3 switch — VLAN trunking, port security	192.168.1.130

Network Topology



R1 serves as the primary gateway and default route originator for production lab networks. R2 handles isolated test networks and peers with R1 over a dedicated link. R3 provides routing from the DMZ 192.168.2.0/24 segment. The 3560X connects physical hosts, trunks VLANs to the Proxmox host, and provides Layer 3 inter-VLAN routing across six VLANs.

All three routers form OSPF adjacencies over 10.30.0.0/29 using MD5 authentication. Passive-interface is set by default on R1; only G0/3 (the P2P link) participates in OSPF hellos. R2 and

R3 follow the same passive-interface pattern, advertising their stub networks into the OSPF domain without sending hellos on end-host-facing interfaces.

Device	Interface	IP Address	Network / Role
r1	G0/0	192.168.1.6/24	Prod_LAN — upstream gateway
r1	G0/1	192.168.100.6/24	Lab_LAN1 — K3s cluster, Docker hosts
r1	G0/2	192.168.200.6/24	Lab_LAN2 — SOC namespace, server-admin
r1	G0/3	10.30.0.1/29	P2P link to R2/R3 — OSPF Area 0
r1	Loopback1	192.168.255.1/32	OSPF router-id / management
r2	G0/0	10.30.0.2/29	P2P uplink to R1 — OSPF Area 0
r2	G0/1	192.168.3.9/24	Isolated test network 1
r2	Loopback1	192.168.255.2/32	OSPF router-id / management
r3	G1	192.168.2.9/24	Lab_Ext / DMZ segment
r3	G3	10.30.0.3/29	P2P uplink to R1 — OSPF Area 0
r3	Loopback1	192.168.255.3/32	OSPF router-id / management
cisco-3560	Vlan10 (Prod_LAN)	192.168.1.130/24	L3 SVI — Prod_LAN
cisco-3560	Vlan20 (DMZ)	192.168.2.130/24	L3 SVI — DMZ
cisco-3560	Vlan30	192.168.3.130/24	L3 SVI — Lab segment
cisco-3560	Vlan100 (Lab_LAN1)	192.168.100.130/24	L3 SVI — Lab_LAN1
cisco-3560	Vlan120 (ISO_LAN)	10.20.0.130/24	L3 SVI — Isolated LAN
cisco-3560	Vlan200 (Lab_LAN2)	192.168.200.130/24	L3 SVI — Lab_LAN2

Security Configuration

Access Control

- SSH v2 only — Telnet disabled on all devices (transport input ssh)
- VTY lines restricted to 192.168.1.0/24 via MGMT_ACCESS ACL
- Enable and user secrets hashed with Type 9 (SCRYPT) on all routers; Type 4 on 3560X (IOS 15.0 limitation)
- service password-encryption enabled on all devices
- HTTP/HTTPS management interface disabled (no ip http server / no ip http secure-server)

OSPF Authentication

- MD5 authentication on all OSPF-enabled interfaces (ip ospf authentication message-digest)
- Shared key configured per interface; passive-interface default prevents OSPF hellos on end-host segments
- R3 uses uRPF (ip verify unicast source reachable-via rx) on uplink interfaces for anti-spoofing

3560X Layer 2 Hardening

- DHCP snooping enabled on VLANs 10, 20, 30, 100, 120, 200

- Rapid-PVST spanning tree with portfast bpduguard default on all access ports
- Port security with sticky MACs on physical device ports (G0/13, G0/17)
- Unused ports assigned to VLAN 999, port-security enabled, administratively shut down
- LACP port-channel (Po1) bonding Proxmox host uplinks (G0/14, G0/16) — DHCP snooping trust on LAG members

Banners

All devices enforce login, exec, and incoming banners identifying the lab domain (shadowitlab.com) and prohibiting unauthorized access.

```
***************************************************
* Homelab, shadowitlab.com                        *
* AUTHORIZED ACCESS ONLY                          *
* Unauthorized access is prohibited and will be   *
* monitored, logged, and reported.                *
***************************************************
```

NetFlow / Flexible NetFlow Configuration

R1, R2 and R3 export Flexible NetFlow v9 to ntopng at 192.168.1.48 UDP/2055. The C3560X does not have NetFlow configured in the current template. Flow records capture source/destination IP, ports, protocol, ToS, direction, and byte/packet counters with 60-second active cache timeouts.

```
! net flow configuration
flow record ntop-record
 description NetFlow record for ntopng
 match ipv4 source address
 match ipv4 destination address
 match transport source-port
 match transport destination-port
 match interface input
 match interface output
 match ipv4 tos
 match flow direction
 match ipv4 protocol
 collect counter bytes long
 collect counter packets long
 collect timestamp sys-uptime first
 collect timestamp sys-uptime last
flow exporter ntop-exporter
 description Export to ntopng
 destination 192.168.1.48
 source GigabitEthernet1
 transport udp 2055
 template data timeout 60
flow monitor ntop-monitor
 description Monitor for ntopng
 exporter ntop-exporter
 cache timeout active 60
 record ntop-record
```

Physical Switch — WS-C3560X-24T-S

Attribute	Value
Model	WS-C3560X-24T-S
IOS Version	15.0(2)SE (C3560E-UNIVERSALK9-M)
License Level	IP Services (Permanent)
Ports	24x GigabitEthernet + 2x Ten GigabitEthernet + 1x FastEthernet (mgmt)
Memory	262144K DRAM
Serial Number	FDO1637P0KD
MAC Address	6C:20:56:9B:81:80
Uplink to Proxmox	LACP Port-Channel (G0/14 + G0/16)
Spanning Tree	Rapid-PVST

Virtual Router — R1 (IOSv 15.9(3)M6)

Attribute	Value

Platform	Cisco IOSv (VIOS-ADVENTERPRISEK9-M) — KVM guest in Proxmox
IOS Version	15.9(3)M6 RELEASE SOFTWARE (fc1)
Interfaces	4x GigabitEthernet

Virtual Router — R2 (IOSv 15.9(3)M6)

Attribute	**Value**
Platform	Cisco IOSv (VIOS-ADVENTERPRISEK9-M) — KVM guest in Proxmox
IOS Version	15.9(3)M6 RELEASE SOFTWARE (fc1)
Interfaces	3x GigabitEthernet

Virtual Router — R3 (CSR1000V)

Attribute	**Value**
Platform	CSR1000V (VXE) — KVM guest in Proxmox
IOS XE Version	16.12.5 (Gibraltar)
License Level	AX (Smart License)
Interfaces	3x GigabitEthernet

Configuration Templates

R1 — Primary Router (vIOS 15.9(3)M6)

```
hostname r1
ip domain name home.com
cdp run
ntp server time.google.com
ntp server time.cloudflare.com
no ip http server
no ip http secure-server
ip name-server 192.168.1.154
ip name-server 192.168.1.153
logging host 192.168.1.178
logging trap informational
logging origin-id hostname
service password-encryption
no ip finger
no ip source-route
username paul privilege 15 secret 9 <hash>
enable secret 9 <hash>
ip ssh version 2
ip ssh server algorithm mac hmac-sha2-256
ip ssh server algorithm encryption aes256-cbc
ip ssh time-out 60
ip ssh authentication-retries 5
ip access-list standard MGMT_ACCESS
 permit 192.168.1.0 0.0.0.255
```

```
line vty 0 4
 login local
 transport input ssh
 access-class MGMT_ACCESS in
interface Loopback1
 ip address 192.168.255.1 255.255.255.255
interface GigabitEthernet0/0
 ip address 192.168.1.6 255.255.255.0
 ip flow monitor ntop-monitor input
 ip flow monitor ntop-monitor output
 duplex full
interface GigabitEthernet0/1
 ip address 192.168.100.6 255.255.255.0
 ip flow monitor ntop-monitor input
 ip flow monitor ntop-monitor output
interface GigabitEthernet0/2
 ip address 192.168.200.6 255.255.255.0
 ip flow monitor ntop-monitor input
 ip flow monitor ntop-monitor output
interface GigabitEthernet0/3
 ip address 10.30.0.1 255.255.255.248
 ip ospf 1 area 0
 ip ospf authentication message-digest
 ip ospf message-digest-key 1 md5 <key>
 ip flow monitor ntop-monitor input
 ip flow monitor ntop-monitor output
router ospf 1
 area 0 authentication message-digest
 passive-interface default
 no passive-interface GigabitEthernet0/3
 network 192.168.1.0 0.0.0.255 area 0
 network 192.168.100.0 0.0.0.255 area 0
 network 192.168.200.0 0.0.0.255 area 0
 network 10.30.0.0 0.0.0.7 area 0
 network 192.168.255.1 0.0.0.0 area 0
 default-information originate
ip route 10.20.0.0 255.255.255.0 192.168.1.1
snmp-server community sh4d0wi7l4b RO
snmp-server location Lab
snmp-server host 192.168.1.178 version 2c <key>
```

R2 — Secondary Router (vIOS 15.9(3)M6)

```
hostname r2
ip domain name home.com
cdp run
ntp server time.google.com
ntp server time.cloudflare.com
no ip http server
no ip http secure-server
ip name-server 192.168.1.154
ip name-server 192.168.1.153
logging host 192.168.1.178
logging trap informational
service password-encryption
no ip source-route
username paul privilege 15 secret 9 <hash>
enable secret 9 <hash>
ip ssh version 2
ip ssh server algorithm mac hmac-sha2-256
ip ssh server algorithm encryption aes256-cbc
ip ssh time-out 60
ip ssh authentication-retries 5
ip access-list standard MGMT_ACCESS
 permit 192.168.1.0 0.0.0.255
line vty 0 4
 login local
 transport input ssh
 access-class MGMT_ACCESS in
!interface Loopback1
 ip address 192.168.255.2 255.255.255.255
interface GigabitEthernet0/0
 ip address 10.30.0.2 255.255.255.248
 ip flow monitor ntop-monitor input
 ip flow monitor ntop-monitor output
 ip ospf authentication message-digest
 ip ospf message-digest-key 1 md5 <key>
 duplex full
interface GigabitEthernet0/1
 ip address 192.168.3.9 255.255.255.0
 ip flow monitor ntop-monitor input
 ip flow monitor ntop-monitor output
router ospf 1
 passive-interface GigabitEthernet0/1
 network 192.168.3.0 0.0.0.255 area 0
 network 10.30.0.0 0.0.0.7 area 0
 network 192.168.255.2 0.0.0.0 area 0
```

R3 — CSR1000V (IOS XE 16.12.5)

```
hostname r3
ip domain name home.com
cdp run
ntp server time.google.com
ntp server time.cloudflare.com
no ip http server
no ip http secure-server
ip name-server 192.168.1.154
ip name-server 192.168.1.153
logging host 192.168.1.178
logging trap informational
no ip source-route
clock timezone EST -5
clock summer-time EDT recurring 2 Sun Mar
2:00 1 Sun Nov 2:00
username paul secret 9 <hash>
enable secret 9 <hash>
ip ssh version 2
ip ssh server algorithm mac hmac-sha2-512
ip ssh server algorithm encryption aes256-ctr
ip ssh time-out 60
ip ssh authentication-retries 5
ip access-list standard MGMT_ACCESS
 permit 192.168.1.0 0.0.0.255
interface Loopback1
 ip address 192.168.255.3 255.255.255.255
interface GigabitEthernet1
 ip address 192.168.2.9 255.255.255.0
 ip flow monitor ntop-monitor input
 ip flow monitor ntop-monitor output
 ip verify unicast source reachable-via rx
 negotiation auto
 cdp enable
interface GigabitEthernet3
 ip address 10.30.0.3 255.255.255.248
 ip ospf 1 area 0
 ip ospf authentication message-digest
 ip ospf message-digest-key 1 md5 <key>
 ip flow monitor ntop-monitor input
 ip flow monitor ntop-monitor output
 negotiation auto
 cdp enable
router ospf 1
 passive-interface GigabitEthernet1
 network 10.30.0.0 0.0.0.7 area 0
 network 192.168.2.0 0.0.0.255 area 0
 network 192.168.255.3 0.0.0.0 area 0
```

Cisco-3560 — WS-C3560X-24T-S (IOS 15.0(2)SE)

```
hostname cisco-3560
ip routing
ip domain-name home.com
no ip http server
no ip http secure-server
ip name-server 192.168.1.154
ip name-server 192.168.1.153
logging host 192.168.1.178
logging console informational
logging origin-id hostname
logging source-interface Vlan10
ntp server time.google.com
ntp server time.cloudflare.com
service password-encryption
spanning-tree mode rapid-pvst
spanning-tree portfast bpduguard default
spanning-tree extend system-id
username paul secret 4 <hash>
enable password 7 <hash>
ip ssh version 2
ip ssh dh min size 2048
ip ssh time-out 60
ip ssh authentication-retries 5
ip dhcp snooping
ip dhcp snooping vlan 10,20,30,100,120,200
ip access-list standard MGMT_ACCESS
 permit 192.168.1.0 0.0.0.255
line vty 0 4
 login local
 transport input ssh
 access-class MGMT_ACCESS in
line vty 5 15
 login local
 transport input ssh
 access-class MGMT_ACCESS in
! --- Port Assignments ---
interface GigabitEthernet0/1
 description Fortigate Prod_LAN intf
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 10
 switchport trunk allowed vlan
10,20,30,100,120,200
 switchport mode trunk
```

```
interface GigabitEthernet0/2
 description Proxmox host Prod_LAN intf
 switchport access vlan 10
 spanning-tree portfast
 spanning-tree bpduguard enable
interface range GigabitEthernet0/3 - 12
 switchport access vlan 10
 switchport mode access
 spanning-tree bpduguard enable
 spanning-tree portfast
 shutdown
interface GigabitEthernet0/13
 switchport access vlan 30
 switchport mode access
 switchport port-security
 switchport port-security violation restrict
 switchport port-security mac-address sticky
 interface GigabitEthernet0/14
 description LACP to Proxmox host
 switchport access vlan 30
 switchport mode access
 channel-group 1 mode active
 ip dhcp snooping trust
interface GigabitEthernet0/15
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 30
 switchport trunk allowed vlan
10,20,30,100,120,200
 switchport mode trunk
interface GigabitEthernet0/16
 description LACP to Proxmox host
 switchport access vlan 30
 switchport mode access
 channel-group 1 mode active
 ip dhcp snooping trust
interface GigabitEthernet0/17
 switchport access vlan 30
 switchport port-security
 switchport port-security violation restrict
 switchport port-security mac-address sticky
interface GigabitEthernet0/21
 description to Fortigate LAN2 intf (Lab_LAN1)
 switchport access vlan 100
 switchport mode access
```

```
interface GigabitEthernet0/24
 description Uplink to Prod_LAN and Asus
Router
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 10
 switchport trunk allowed vlan
10,20,30,100,120,200
 switchport mode trunk
 switchport nonegotiate
 ip dhcp snooping trust
interface Port-channel1
 description Proxmox host LAG
 switchport access vlan 30
 switchport mode access
 ip dhcp snooping trust
! --- SVIs ---
interface Vlan1
 no ip address
 shutdown
interface Vlan10
 description Prod_LAN
 ip address 192.168.1.130 255.255.255.0
 ip flow monitor ntop-monitor input
interface Vlan20
 description DMZ
 ip address 192.168.2.130 255.255.255.0
interface Vlan30
 ip address 192.168.3.130 255.255.255.0
 ip flow monitor ntop-monitor input
interface Vlan100
 description Lab_LAN1
 ip address 192.168.100.130 255.255.255.0
interface Vlan120
 description ISO_LAN
 ip address 10.20.0.130 255.255.255.0
interface Vlan200
 description Lab_LAN2
 ip address 192.168.200.130 255.255.255.0
```

Container Orchestration Architecture

Deployment Overview

The container orchestration layer consists of two complementary platforms: a multi-engine Docker deployment for lightweight, isolated workloads and a dual-node K3s Kubernetes cluster for cloud-native, scalable applications. Docker engines provide simple, isolated runtime environments ideal for single-purpose services, while K3s delivers a fully compliant Kubernetes distribution optimized for low-resource environments. Portainer provides centralized management across both platforms, enabling unified visibility, lifecycle control, and operational consistency. Together, these systems form a hybrid container ecosystem that mirrors modern enterprise architectures where Docker and Kubernetes coexist to support diverse workloads.

Security Impact

- Workload isolation across six independent Docker engines reduces blast radius
- Kubernetes network policies and ingress controls enforce strict east-west and north-south traffic boundaries
- Portainer centralizes access control and auditability across all container platforms
- K3s certificate automation ensures secure service-to-service communication
- Segmented runtimes prevent cross-container compromise
- MetalLB and NGINX Ingress enforce controlled exposure of internal services

Deployment Rationale

Enterprises frequently operate hybrid container environments where Docker supports lightweight services and Kubernetes orchestrates scalable, distributed applications. This architecture demonstrates proficiency with both paradigms—multi-engine Docker for simplicity and K3s for cloud-native orchestration. The deployment mirrors real-world operational patterns including ingress management, persistent storage provisioning, network policy enforcement, and centralized container lifecycle management.

Architecture Principles Alignment

Defense in Depth: Multiple container runtimes, isolated engines, Kubernetes network policies, and ingress controls

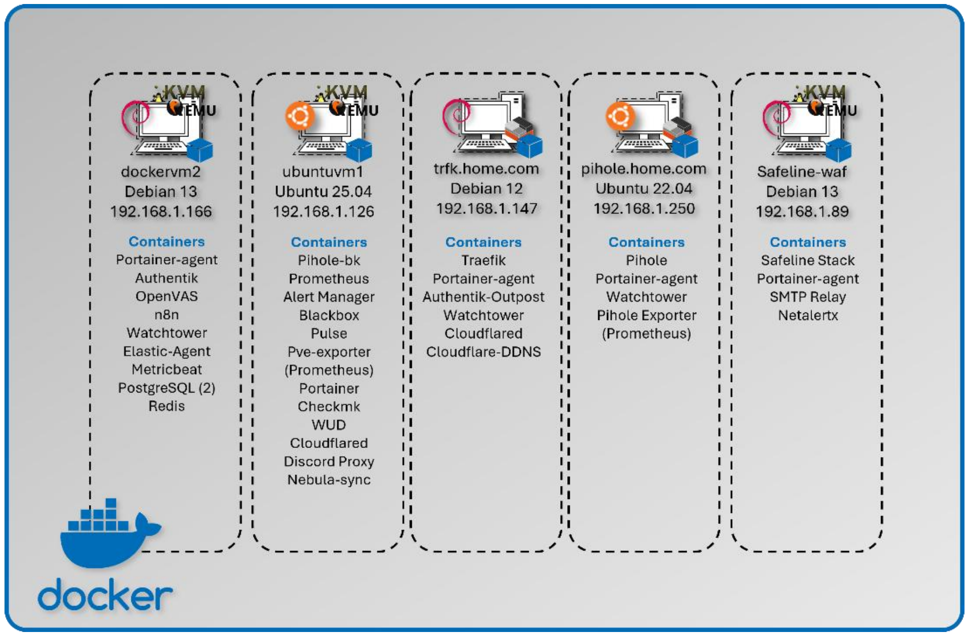
Secure by Design: TLS-enabled Kubernetes control plane, Portainer RBAC, minimal host dependencies

Zero Trust: No container or service implicitly trusted; identity, ingress, and network policies enforced continuously

Multi-Engine Docker Deployment

Deployment Overview

The lab operates five independent Docker engines, each hosting isolated workloads to minimize cross-service impact. Portainer Community Edition provides a centralized GUI for managing all engines, while Portainer Agents enable secure API communication with remote hosts.



Security Impact

- Independent Docker engines prevent lateral movement between workloads
- Portainer RBAC restricts administrative access to container hosts
- Engine-level isolation reduces the impact of misconfigurations or compromised containers
- Minimal shared dependencies reduce systemic risk

Deployment Rationale

Multi-engine Docker deployments mirror enterprise environments where isolated runtimes support operational segmentation, compliance boundaries, or workload separation. This design demonstrates proficiency with distributed container management, remote engine control, and secure API-driven orchestration.

Architecture Principles Alignment

Defense in Depth: Multiple engines create natural segmentation boundaries

Secure by Design: Portainer Agents use secure API channels; Home Assistant isolated for safety

Zero Trust: No container shares trust with another engine; all access controlled centrally

Configuration

Engine Location	Purpose	Key Containers
Ubuntu VM (DockerVM1)	Central management & monitoring	Portainer; Prometheus; Checkmk; WUD; Pulse; Cloudflared
Debian VM (DockerVM2)	Identity & security services	Authentik; OpenVAS; PostgreSQL; Elastic Agent; n8n
LXC-1 (Primary DNS)	Network infrastructure	Pi-hole; prometheus-exporter
LXC-2 (Ingress)	Reverse proxy & SSO	Traefik; Authentik outpost
Safeline-WAF	Web Application Firewall and Gmail notifications	Safeline stack, SMTP relay

Docker Compose

Compose files are created in VS Code and stored in a Github repository for version control.

Lab Configs > Docker Compose > docker-compose_pihole.yml > ...

```
version: "3.9"

▷Run All Services
services:
  ▷Run Service
  pihole:
    image: pihole/pihole:latest
    container_name: pihole
    hostname: pihole
    restart: unless-stopped

    ports:
      - "53:53/tcp"
      - "53:53/udp"
      - "80:80/tcp"
      - "443:443/tcp"

    dns:
      - 192.168.1.250
      - 192.168.1.126

    environment:
      TZ: "America/New_York"
      WEBPASSWORD: "paul"
      DNSMASQ_USER: "pihole"

    volumes:
      - /opt/pihole/etc-pihole:/etc/pihole
      - /opt/pihole/etc-dnsmasq.d:/etc/dnsmasq.d

    healthcheck:
      test: ["CMD-SHELL", "curl -sS http://localh
      interval: 30s
      timeout: 5s
      retries: 5
      start_period: 20s
```

Lab Configs > Docker Compose > docker-compose_authentik_stack.yml > {} volumes > {} redis > driver

```
▷Run All Services
services:
  ▷Run Service
  postgresql:
    image: docker.io/library/postgres:16-alpine
    restart: unless-stopped
    healthcheck:
      test: ["CMD-SHELL", "pg_isready -d $${POSTGRES_DB} -U $${POSTGRES_USE
      start_period: 20s
      interval: 30s
      retries: 5
      timeout: 5s
    volumes:
      - database:/var/lib/postgresql/data
    environment:
      POSTGRES_PASSWORD: D@ll@sc0wb0ys
      POSTGRES_USER: admin
      POSTGRES_DB: authentik-db

  ▷Run Service
  redis:
    image: docker.io/library/redis:alpine

    command: --save 60 1 --loglevel warning
    restart: unless-stopped
    healthcheck:
      test: ["CMD-SHELL", "redis-cli ping | grep PONG"]
      start_period: 20s
      interval: 30s
      retries: 5
      timeout: 3s
    volumes:
      - redis:/data
  ▷Run Service
  server:
    image: ${AUTHENTIK_IMAGE:-ghcr.io/goauthentik/server}:${AUTHENTIK_TAG:-
    restart: unless-stopped

    command: server
    environment:
      AUTHENTIK_SECRET_KEY: [redacted]
      AUTHENTIK_REDIS__HOST: redis
      AUTHENTIK_POSTGRESQL__HOST: postgresql
      AUTHENTIK_POSTGRESQL__USER: admin
      AUTHENTIK_POSTGRESQL__NAME: authentik-db
      AUTHENTIK_POSTGRESQL__PASSWORD: D@ll@sc0wb0ys
      AUTHENTIK_ERROR_REPORTING__ENABLED: true
    volumes:
      - ./media:/media
      - ./custom-templates:/templates
      - /opt/authentik/certs/:/certs/

    ports:
      - "80:9000"
      - "443:9443"
    depends_on:
      postgresql:
        condition: service_healthy
```

Cloud-Native Kubernetes Cluster Deployment

Deployment Overview

The lab runs a dual-node K3s cluster (one control plane, one worker) on Red Hat Enterprise Linux 10 VMs within the 192.168.200.0/24 subnet. K3s is a fully compliant Kubernetes distribution packaged as a single binary, minimizing external dependencies and simplifying cluster operations. It includes a batteries-included stack: containerd runtime, Flannel CNI, CoreDNS, Kube-router, Local-path-provisioner, and Spegel registry mirror.

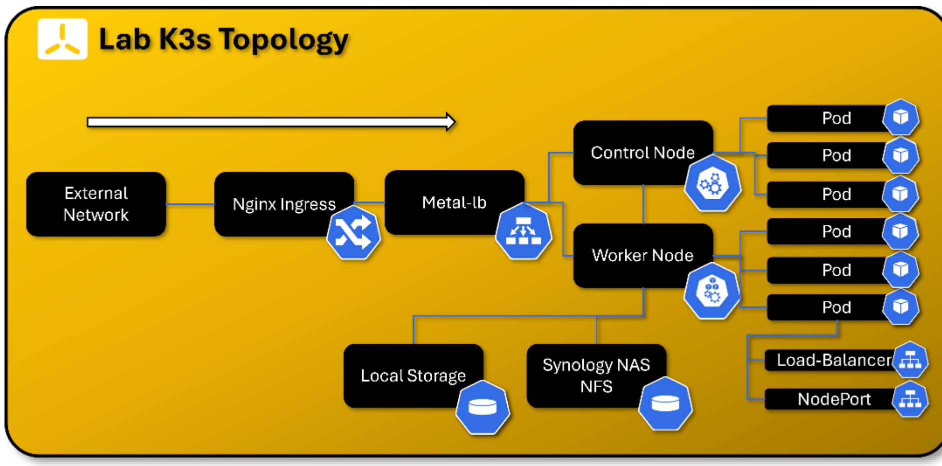
The embedded Traefik ingress controller and ServiceLB load balancer have been intentionally disabled and replaced with NGINX Ingress and MetalLB, providing enterprise-grade ingress routing and Layer-2 external IP allocation. Portainer integrates with the cluster via a DaemonSet-based Portainer Agent for full lifecycle management.

Security Impact

- K3s certificate automation secures all control plane and node communications
- NGINX Ingress enforces controlled north-south traffic with TLS termination
- MetalLB provides predictable, controlled external IP allocation
- Flannel CNI and Kube-router enforce network segmentation and policy enforcement
- Minimal external dependencies reduce attack surface
- Local-path-provisioner isolates persistent volumes per workload

Deployment Rationale

K3s is ideal for homelab and edge environments requiring full Kubernetes functionality with reduced operational overhead. This deployment demonstrates proficiency with Kubernetes networking, ingress management, persistent storage provisioning, and cluster lifecycle operations. Replacing Traefik and ServiceLB with NGINX Ingress and MetalLB mirrors enterprise-grade ingress and load-balancing patterns.

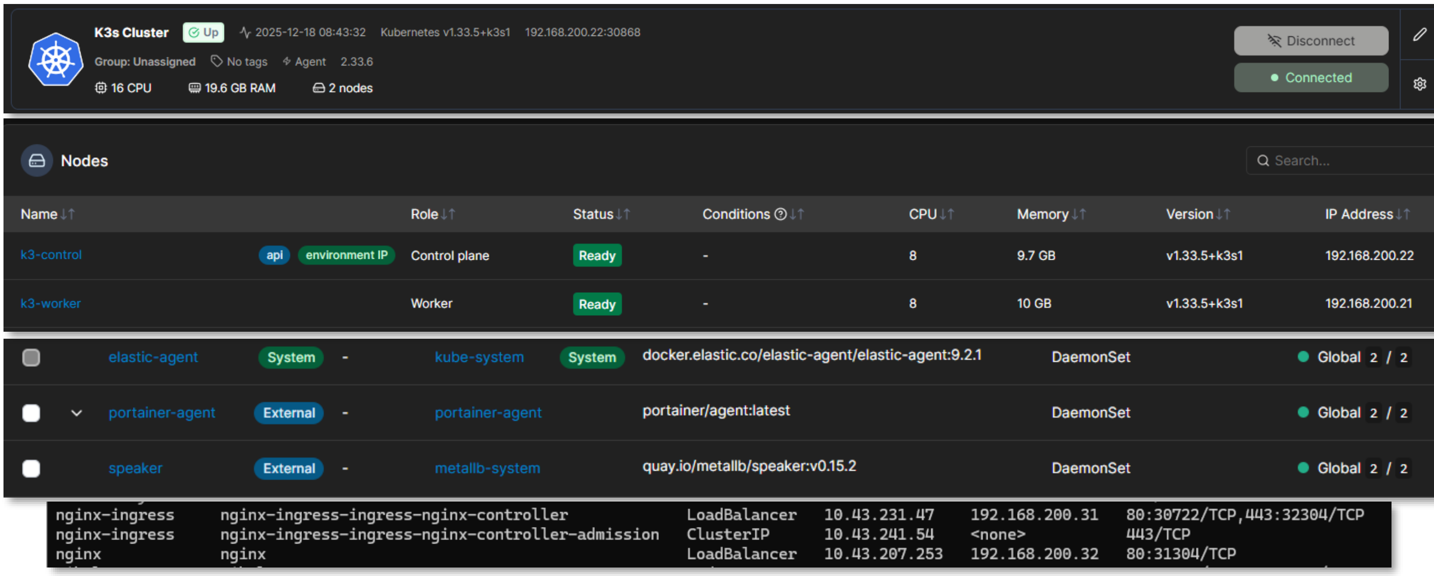


Architecture Principles Alignment

Defense in Depth: Network policies, ingress controls, and runtime isolation across pods and namespaces

Secure by Design: TLS-secured control plane, minimal dependencies, automated certificate rotation

Zero Trust: Every pod, service, and ingress request authenticated and authorized; no implicit trust between namespaces



Core Infrastructure Services:

Component	Namespace	Purpose	Deployment Type
MetalLB Controller	metallb-system	Layer 2/BGP load balancer controller for bare-metal Kubernetes	Deployment (1 replica)
MetalLB Speaker	metallb-system	Announces LoadBalancer IPs via ARP/BGP	DaemonSet (runs on all nodes)
Nginx Ingress Controller	nginx-ingress	HTTP/HTTPS ingress traffic routing and SSL termination	Helm chart DaemonSet (runs on all nodes)
Portainer Agent	portainer-agent	Cluster management and monitoring via Portainer UI	DaemonSet (runs on all nodes)
Cert-Manager	Cert-manager	StepCA/ACME client for nginx-ingress SSL termination	Helm Chart Deployment

MetalLB Load Balancer Configuration:

- **Address Pool**: 192.168.200.30-192.168.200.49 (20 available IPs)
- **Mode**: Layer 2 (ARP-based)
- **Purpose**: Provides external IPs for LoadBalancer-type services in bare-metal environment

Ingress Architecture:

- **Controller**: Nginx Ingress Controller (official Kubernetes ingress-nginx)
- **External Access**: LoadBalancer service at 192.168.200.31 (HTTP: 80, HTTPS: 443)
- **Use Case**: Centralized ingress point for HTTP-based services with path-based routing. Provides rate limiting, IP allow list and TLS/HTTPS connectivity to the NGINX Webserver. Certificates are delivered via Cert-Manager service that is auto-generated from StepCA/ACME.

Nginx Namespace - Web Services

Application	Type	Image	Replicas	External IP	Ports	Purpose
nginx	Deployment	linuxserver/nginx:latest	2/2	Ingress IP	80 (HTTP)	Static web server for lab documentation and demos

Infrastructure Namespaces

metallb-system Namespace:

Application	Type	Replicas	Purpose
controller	Deployment	1/1	MetalLB controller managing IP address allocation
speaker	DaemonSet	2/2	MetalLB speaker announcing IPs via Layer 2 (ARP)

nginx-ingress Namespace:

Application	Type	Replicas	External IP	Purpose
nginx-ingress-controller	DaemonSet	2/2	192.168.200.31	HTTP/HTTPS ingress controller for path-based routing

portainer-agent Namespace:

Application	Type	Replicas	Purpose
portainer-agent	DaemonSet	2/2	Portainer agent for cluster management via Portainer UI

Cert-manager Namespace:

Application	Type	Replicas	External IP	Purpose
Cert-Manager	Helm	1/1	-	StepCA/ACME client for nginx-ingress SSL termination
cert-manager-	Helm	1/1	-	
cert-manager-webhook	Helm	1/1	-	

SOC Namespace - Security Operations Center Platform

The SOC namespace hosts the lab's comprehensive Security Operations Center platform, implementing a modern threat intelligence, incident response, and security orchestration architecture. Integrating threat intelligence sharing (MISP), automated analysis (Cortex), workflow automation (Shuffle), and case management (TheHive).

Architecture Overview

Namespace Purpose*: Centralized security operations platform providing end-to-end incident response capabilities from threat intelligence ingestion through automated workflow execution, observable analysis, case resolution, and IOC sharing.*

Design Rationale:

- ***Unified Security Platform****: Single namespace for all SOC functions enables tight integration and simplified network policies*
- ***Workflow Automation****: Shuffle SOAR orchestrates complex multi-tool workflows without requiring custom code development*
- ***Scalability****: Kubernetes orchestration allows horizontal scaling of analysis workers (Cortex, Shuffle Orborus) during high-volume incident periods*
- ***Resilience****: StatefulSets ensure data persistence for critical components (Cassandra, Elasticsearch, OpenSearch)*
- ***Integration****: Native Kubernetes networking facilitates service-to-service communication with other lab security tools*

SOC Namespace - Security Operations Center

Application	Type	Image	Replicas	External IP	Ports	Purpose

thehive	Helm	strangebee/thehive:5.5.13-1	1/1	192.168.200.33	9000 (HTTP), 9095 (metrics)	Security incident response platform and case management
cortex	Deployment	thehiveproject/cortex:latest	1/1	192.168.200.40	9001 (HTTP)	Observable analysis engine with automated responders
cassandra	StatefulSet	cassandra:4.1.7	1/1	192.168.200.36	9042 (CQL)	Distributed database for TheHive persistent storage
elasticsearch	StatefulSet	elasticsearch:9.2.2	1/1	192.168.200.34	9200 (HTTP)	Search and analytics engine for cases and observables
misp-core	Deployment	misp-docker/misp-core:latest	1/1	192.168.200.37	80/443 (HTTP/HTTPS)	Threat intelligence platform and IOC management
misp-db	Deployment	mariadb:10.11	1/1	ClusterIP only	3306 (MySQL)	MySQL database for MISP data
misp-redis	Deployment	valkey/valkey:7.2	1/1	ClusterIP only	6379 (Redis)	Redis cache for MISP sessions and jobs
misp-modules	Deployment	misp-docker/misp-modules:latest	1/1	ClusterIP only	6666 (HTTP)	MISP enrichment and expansion modules
misp-guard	Deployment	misp-docker/misp-guard:latest	1/1	ClusterIP only	8888 (HTTP)	Security proxy for MISP core protection
misp-mail	Deployment	egos-tech/smtp:latest	1/1	192.168.200.38	25 (SMTP)	SMTP relay for threat intelligence email sharing
msmtp-relay	Deployment	alpine:latest	1/1	ClusterIP only	N/A	Lightweight SMTP relay for internal notifications
shuffle-frontend	Deployment	shuffle-frontend:latest	1/1	192.168.200.41	80/443 (HTTP/HTTPS)	React-based web UI for visual workflow design, execution monitoring, and SOAR administration
Shuffle-backend	Deployment	shuffle-backend:latest	1/1	ClusterIP only	5001	Go-based backend API handling workflow orchestration, webhook processing, and app management
Shuffle-opensearch	StatefulSet	opensearch:3.2.0	1/1	ClusterIP only	9200 (HTTP)	Search and analytics engine for workflow definitions, execution history, and audit logs
Shuffle-orborus	Deployment	shuffle-orborus:latest	1/1	none		Worker orchestration daemon managing Docker containers for workflow app execution

Server-Admin Namespace - Infrastructure Management

Application	Type	Image	Replicas	External IP	Ports	Purpose
patchmon-frontend	Deployment	patchmon-frontend:latest	1/1	192.168.200.35	3000 (HTTP)	Web UI for Windows patch management dashboard
patchmon-backend	Deployment	patchmon-backend:latest	1/1	192.168.200.39	3001 (HTTP API)	Backend API for patch compliance tracking
patchmon-database	Deployment	postgres:17-alpine	1/1	ClusterIP only	5432 (PostgreSQL)	PostgreSQL database for patch status data
patchmon-redis	Deployment	redis:7-alpine	1/1	ClusterIP only	6379 (Redis)	Redis cache for session management

Network Services Summary

LoadBalancer Services (Externally Accessible)

Service Name	Namespace	Type	External IP	Ports	Application	Access Method
nginx-ingress-controller	nginx-ingress	LoadBalancer	192.168.200.31	80, 443	Ingress Controller	Primary HTTP/HTTPS ingress point
nginx	nginx	LoadBalancer	192.168.200.32	80	Nginx Web Server	Static web content hosting
thehive	soc	LoadBalancer	192.168.200.33	9000, 9095	TheHive SIRP	Case management UI
elasticsearch	soc	LoadBalancer	192.168.200.34	9200	Elasticsearch	Search API (admin only)
patchmon-frontend	server-admin	LoadBalancer	192.168.200.35	3000	PatchMon UI	Patch management dashboard
cassandra	soc	LoadBalancer	192.168.200.36	9042	Cassandra DB	Database access (admin only)
misp-core	soc	LoadBalancer	192.168.200.37	80, 443	MISP Platform	Threat intelligence portal
misp-mail	soc	LoadBalancer	192.168.200.38	25	SMTP Relay	Email-based threat sharing
patchmon-backend	server-admin	LoadBalancer	192.168.200.39	3001	PatchMon API	Backend API (internal)
cortex	soc	LoadBalancer	192.168.200.40	9001	Cortex Engine	Observable analysis API
Shuffle	Soc	LoadBalancer	192.168.200.41	80/443	Shuffle	Automation UI

Access Control:

- All services accessible only via Tailscale VPN (no direct internet exposure)
- Administrative services (Elasticsearch, Cassandra) require Authentik SSO + MFA
- Network policies enforce namespace isolation and least-privilege access
- pfSense firewall rules restrict access by source IP and service port

ClusterIP Services (Internal Only)

Service Name	Namespace	Cluster IP	Ports	Purpose
kubernetes	default	10.43.0.1	6443	Kubernetes API server
patchmon-database	server-admin	10.43.3.202	5432	PostgreSQL backend for PatchMon
misp-db	soc	10.43.151.59	3306	MariaDB backend for MISP
misp-redis	soc	10.43.131.214	6379	Redis cache for MISP
misp-modules	soc	10.43.234.246	6666	MISP enrichment modules
misp-guard	soc	10.43.97.195	8888	MISP security proxy
patchmon-backend	server-admin	10.43.99.91	3001	PatchMon API (internal routing)
patchmon-redis	server-admin	10.43.126.102	6379	Redis cache for PatchMon
metallb-webhook-service	metallb-system	10.43.122.116	9443	MetalLB webhook validation
shuffle-backend	soc	10.43.189.138	5001	Shuffle backend integration
shuffle-opensearch	soc	10.43.137.30	9200	Shuffle database
nginx-ingress-admission	nginx-ingress	10.43.241.54	443	Ingress admission webhook
cert-manager-webhook	cert-manager	10.43.150.38	443	SSL termination for nginx-ingress

Network Policies:

- Default deny all ingress/egress traffic
- Explicit allow rules for required service-to-service communication
- Database services (PostgreSQL, MariaDB, Redis) accessible only from their respective application pods
- Rate limiting for Ingress HTTP services
- SSL termination for all ingress HTTP services

Example Workload: Pi-hole DNS

- Deployment: 2 replicas with anti-affinity rules
- Persistent Storage: local-path-provisioner (hostPath-based PVCs)
- Exposure: MetalLB LoadBalancer with external IP
- Health Checks: HTTP liveness/readiness probes on /admin
- Resource Limits: 512MB memory, 0.5 CPU per pod
- Persistent Volume Claims / Volumes: pihole-config and dnsmasq-config

be Manifests > ! pihole-full.yml > {} spec > {} template > {} spec > [ ] containers > {} 0

You, yesterday | 1 author (You) | io.k8s.api.core.v1.Service (v1@service.json) | io.k8s.api.app

```yaml
apiVersion: v1
kind: Namespace
metadata:
  name: pihole
---
apiVersion: v1
kind: PersistentVolumeClaim
metadata:
  name: pihole-config
  namespace: pihole
spec:
  accessModes:
    - ReadWriteOnce
  resources:
    requests:
      storage: 500Mi
---
apiVersion: v1
kind: PersistentVolumeClaim
metadata:
  name: dnsmasq-config
  namespace: pihole
spec:
  accessModes:
    - ReadWriteOnce
  resources:
    requests:
      storage: 500Mi
---
apiVersion: apps/v1
kind: Deployment
metadata:
  name: pihole
  namespace: pihole
spec:
  replicas: 2
  selector:
    matchLabels:
      app: pihole
  template:
    metadata:
      labels:
        app: pihole
    spec:
      dnsPolicy: None
      dnsConfig:
        nameservers:
          - 192.168.1.250
          - 192.168.1.126
```

```yaml
---
apiVersion: v1
kind: Service
metadata:
  name: pihole
  namespace: pihole
spec:
  type: LoadBalancer
  loadBalancerIP: 192.168.200.33  #
  selector:
    app: pihole
  ports:
    - name: dns-tcp
      protocol: TCP
      port: 53
      targetPort: 53
    - name: dns-udp
      protocol: UDP
      port: 53
      targetPort: 53
    - name: http
      protocol: TCP
      port: 80
      targetPort: 80
    - name: https
      protocol: TCP
      port: 443
      targetPort: 443
```

```yaml
containers:
  - name: pihole
    image: pihole/pihole:latest
    resources:
      requests:
        cpu: "100m"
        memory: "128Mi"
      limits:        You, 2 seconds ago • Uncommitted
        cpu: "500m"
        memory: "512Mi"
    ports:
      - containerPort: 53
        protocol: TCP
      - containerPort: 53
        protocol: UDP
      - containerPort: 80
        protocol: TCP
      - containerPort: 443
        protocol: TCP
    env:
      - name: TZ
        value: "America/New_York"
      - name: FTLCONF_webserver_api_password
        value: ""
      - name: DNSMASQ_USER
        value: "pihole"
      - name: FTLCONF_dns_listeningMode
        value: "all"
      - name: FTLCONF_dns_upstreams
        value: "192.168.1.252#5335"
    volumeMounts:
      - name: pihole-config
        mountPath: /etc/pihole
      - name: dnsmasq-config
        mountPath: /etc/dnsmasq.d
    readinessProbe:
      httpGet:
        path: /admin/
        port: 80
      initialDelaySeconds: 20
      periodSeconds: 30
      timeoutSeconds: 5
      failureThreshold: 5
volumes:
  - name: pihole-config
    persistentVolumeClaim:
      claimName: pihole-config
  - name: dnsmasq-config
    persistentVolumeClaim:
      claimName: dnsmasq-config
```

Version Control Strategy

- Repository: GitHub repo (infrastructure-as-code)
- Structure: Organized by service (docker-compose/, k8s-manifests/, terraform/Ansible)
- Tooling: VS Code with Docker, Kubernetes, SSH, Ansible, Terraform extensions
- Automation: Watchtower monitors container images, WUD provides update alerts

Appendices

Lab Network Topology, Routing and Domain Namespace

This section outlines the IP addressing scheme, subnet allocations, and internal domain naming conventions used across the Lab infrastructure. It supports modular service deployment, secure DNS resolution, and PKI integration.

IPv4 Addressing , VLAN and Subnet Allocation

Subnet	CIDR	VID	Purpose	Notes
192.168.1.0	/24	10	Production Network	Hosts production environment. Lab services being migrated to lab subnets.
192.168.2.0	/24	20	External LAN / DMZ	DMZ segment. PA-FW DMZ-ISO ethernet1/2 (192.168.2.138) is the NGFW gateway for this segment.
192.168.3.0	/24	30	Isolated Link / MGMT	Port-based VLAN between FortiGate, OfficePC, and Proxmox host.
192.168.100.0	/24	100	Primary Lab LAN	Hosts lab containers, VMs, and services.
192.168.200.0	/24	200	Secondary Lab LAN	Isolated test environments, ephemeral services.
10.20.0.0	/24	120	Isolated LAN (ISO_LAN1)	PA-FW DMZ-ISO inside interface (ethernet1/1, 10.20.0.138). NGFW-enforced segment.
10.30.0.0	/29	130	Cisco Router P2P + PA OSPF	Router links between Cisco R1/R2/R3 and PA-FW DMZ-ISO OSPF interface (10.30.0.4/29).
10.40.0.0	/24	140	PA Site2 Local LAN	Simulated remote branch site. PA-FW Site2 inside interface (ethernet1/1, 10.40.0.2/24). Reachable via IPSec tunnel from ISO_LAN1.
10.10.0.0	/24	110	HA Sync	Isolated subnet for pfSense HA sync/CARP.
192.168.255.0	/24	—	Loopback / Router-ID	OSPF router-IDs and IPSec tunnel endpoints. R1: .1, R2: .2, R3: .3, PA-FW DMZ-ISO: .138, PA-FW Site2: .139.
192.168.99.0	/24	99	OOB Management	Out-of-band management for PA-FW DMZ-ISO (.138) and PA-FW Site2 (.139).
172.31.0.0	/16	—	AWS VPC	AWS us-east-2 default VPC - Reachable via Tailscale; aws-ec2-host1 subnet router
10.128.0.0	/16	—	GCP VPC	GCP us-central1 default VPC - Reachable via Tailscale; gcp-debian-host1 subnet router
10.130.0.0	/16	—	Azure VNet	Azure North Central US homelab-vnet2 - Reachable via Tailscale; az-ubuntu subnet router
100.64.0.0	/10	—	Tailscale CGNAT	Tailscale overlay network - All enrolled nodes; bidirectional mesh

All subnets are routed internally and firewalled to enforce least privilege access between zones.

Routing tables

ASUS Router (192.168.1.1)

Role: Primary gateway for the entire network **Directly connected:** 192.168.1.0/24

Destination Subnet	Next Hop	Notes
192.168.100.0/24	192.168.1.254 (pfSense)	LAB_LAN1
192.168.200.0/24	192.168.1.254 (pfSense)	LAB_LAN2
192.168.2.0/24	192.168.1.254 (pfSense)	EXT_LAN
10.10.0.0/24	192.168.1.254 (pfSense)	Internal VLAN
10.20.0.0/24	192.168.1.138 (PA-FW DMZ-ISO)	ISO_LAN1 via PA-FW
192.168.2.0/24	192.168.1.138 (PA-FW DMZ-ISO)	DMZ via PA-FW
192.168.3.0/24	192.168.1.99 (FortiGate)	ISO segment #2
10.30.0.0/29	192.168.1.6 (Cisco R1)	Cisco P2P + PA OSPF segment
10.40.0.0/24	192.168.1.138 (PA-FW DMZ-ISO)	Site2 remote branch via IPSec
0.0.0.0/0	ISP	Internet
172.31.0.0/16	192.168.1.254 (pfSense)	AWS VPC via Tailscale
10.128.0.0/16	192.168.1.254 (pfSense)	GCP VPC via Tailscale
10.130.0.0/16	192.168.1.254 (pfSense)	Azure VNet via Tailscale
100.64.0.0/10	192.168.1.254 (pfSense)	Tailscale CGNAT range

pfSense HA Firewalls (192.168.1.254 VIP)

Role: Core router for all internal VLANs **Directly connected:** 192.168.1.0/24, 192.168.100.0/24, 192.168.200.0/24, 192.168.2.0/24, 10.10.0.0/24

Destination Subnet	Next Hop	Notes
192.168.1.0/24	Direct	Prod_LAN
192.168.100.0/24	Direct	LAB_LAN1
192.168.200.0/24	Direct	LAB_LAN2
192.168.2.0/24	Direct	EXT_LAN
10.10.0.0/24	Direct	Internal VLAN
10.20.0.0/24	192.168.1.138 (PA-FW DMZ-ISO)	ISO_LAN1 via PA-FW
192.168.3.0/24	192.168.1.99 (FortiGate)	ISO #2
10.30.0.0/29	192.168.1.6 (Cisco R1)	Cisco P2P + PA OSPF segment
10.40.0.0/24	192.168.1.138 (PA-FW DMZ-ISO)	Site2 remote branch via IPSec
0.0.0.0/0	192.168.1.1 (ASUS)	Internet
172.31.0.0/16	Tailscale (TSCALE IF)	AWS VPC via Tailscale subnet router
10.128.0.0/16	Tailscale (TSCALE IF)	GCP VPC via Tailscale subnet router
10.130.0.0/16	Tailscale (TSCALE IF)	Azure VNet via Tailscale subnet router

OPNsense (192.168.1.201)

Role: ISO router for 10.20.0.0/24 **Directly connected:** 10.20.0.0/24 (ISO_LAN), 192.168.1.0/24 (Prod_LAN)

Destination Subnet	Next Hop	Notes
10.20.0.0/24	Direct	ISO_LAN
192.168.1.0/24	Direct	Prod_LAN
192.168.100.0/24	192.168.1.254 (pfSense)	LAB_LAN1
192.168.200.0/24	192.168.1.254 (pfSense)	LAB_LAN2
192.168.2.0/24	192.168.1.254 (pfSense)	EXT_LAN
10.10.0.0/24	192.168.1.254 (pfSense)	Internal VLAN
192.168.3.0/24	192.168.1.99 (FortiGate)	ISO #2
10.30.0.0/29	192.168.1.6 (Cisco R1)	Cisco P2P + PA OSPF segment
10.40.0.0/24	192.168.1.138 (PA-FW DMZ-ISO)	Site2 remote branch via IPSec
0.0.0.0/0	192.168.1.1 (ASUS)	Internet

FortiGate (192.168.1.99)
Role: ISO router for 192.168.3.0/24 **Directly connected:** 192.168.3.0/24 (ISO_LAN), 192.168.1.0/24 (Prod_LAN)

Destination Subnet	Next Hop	Notes
192.168.3.0/24	Direct	ISO_LAN
192.168.1.0/24	Direct	Prod_LAN
192.168.100.0/24	192.168.1.254 (pfSense)	LAB_LAN1
192.168.200.0/24	192.168.1.254 (pfSense)	LAB_LAN2
192.168.2.0/24	192.168.1.254 (pfSense)	EXT_LAN
10.10.0.0/24	192.168.1.254 (pfSense)	Internal VLAN
10.20.0.0/24	192.168.1.138 (PA-FW DMZ-ISO)	ISO_LAN1 via PA-FW
10.30.0.0/29	192.168.1.6 (Cisco R1)	Cisco P2P + PA OSPF segment
10.40.0.0/24	192.168.1.138 (PA-FW DMZ-ISO)	Site2 remote branch via IPSec
0.0.0.0/0	192.168.1.1 (ASUS)	Internet

PA-FW DMZ-ISO (192.168.1.138)
Hostname: pa-fw-dmz.home.com

Role: NGFW for DMZ, ISO_LAN1, and Prod_LAN. IPSec hub to Site2. OSPF peer to Cisco R1/R2/R3.

Directly connected: 192.168.1.0/24 (Prod_LAN, eth1/3), 192.168.2.0/24 (DMZ, eth1/2), 10.20.0.0/24 (ISO_LAN1, eth1/1), 10.30.0.4/29 (OSPF, eth1/4), 192.168.255.138/32 (Lo1 / IPSec source)

OSPF adjacencies: R1 (10.30.0.1), R2 (10.30.0.2), R3 (10.30.0.3) — Area 0 via ISO-DMZ virtual router

Destination Subnet	Next Hop	Source	Notes
192.168.1.0/24	Direct	ethernet1/3	Prod_LAN
192.168.2.0/24	Direct	ethernet1/2	DMZ
10.20.0.0/24	Direct	ethernet1/1	ISO_LAN1
10.30.0.4/29	Direct	ethernet1/4	OSPF peering segment
192.168.255.138/32	Direct	loopback.1	IPSec tunnel source / router-ID
10.40.0.0/24	—	tunnel.40	Site2 via IPSec (static)
192.168.100.0/24	10.30.0.1	OSPF via R1	LAB_LAN1
192.168.200.0/24	10.30.0.1	OSPF via R1	LAB_LAN2
192.168.3.0/24	10.30.0.2	OSPF via R2	ISO_LAN2
192.168.255.1/32	10.30.0.1	OSPF via R1	R1 loopback
192.168.255.2/32	10.30.0.2	OSPF via R2	R2 loopback
192.168.255.3/32	10.30.0.3	OSPF via R3	R3 loopback
192.168.255.139/32	192.168.1.139	Static	PA-FW Site2 loopback (IPSec peer)
0.0.0.0/0	192.168.1.1	Static	Default via ASUS

PA-FW Site2 (192.168.1.139)
Hostname: palo-fw.home.com

Role: Simulated remote branch firewall. IPSec spoke to PA-FW DMZ-ISO.

Directly connected: 10.40.0.0/24 (site2-local, eth1/1), 192.168.1.0/24 (Prod_LAN/uplink, eth1/2), 192.168.255.139/32 (Lo1 / IPSec source)

Destination Subnet	Next Hop	Source	Notes
10.40.0.0/24	Direct	ethernet1/1	Site2 local LAN
192.168.1.0/24	Direct	ethernet1/2	Prod_LAN uplink
192.168.255.139/32	Direct	loopback.1	IPSec tunnel source
10.20.0.0/24	—	tunnel.40	ISO_LAN1 via IPSec to PA-FW DMZ-ISO

0.0.0.0/0	192.168.1.1	Static	Default via ASUS

Cisco r1 vRouter (192.168.1.6)
Hostname: R1.home.com

Role: Core router for Prod, LAB1, LAB2. OSPF DR for 10.30.0.0/29. Default route originator.

Directly connected: 192.168.1.0/24, 192.168.100.0/24, 192.168.200.0/24, 10.30.0.0/29, 192.168.255.1/32 (Lo1)

OSPF adjacencies: R2 (10.30.0.2), R3 (10.30.0.3), PA-FW DMZ-ISO (10.30.0.4) — Area 0

Destination Subnet	Next Hop	Notes
192.168.1.0/24	Direct	Prod_LAN
192.168.100.0/24	Direct	LAB_LAN1
192.168.200.0/24	Direct	LAB_LAN2
10.30.0.0/29	Direct	P2P to R2, R3, PA-FW
192.168.255.1/32	Direct	Lo1
192.168.2.0/24	10.30.0.3	OSPF via R3
192.168.3.0/24	10.30.0.2	OSPF via R2
192.168.255.2/32	10.30.0.2	OSPF via R2
192.168.255.3/32	10.30.0.3	OSPF via R3
10.20.0.0/24	10.30.0.4	OSPF via PA-FW DMZ-ISO
192.168.2.0/24	10.30.0.4	OSPF via PA-FW DMZ-ISO (PA advertisement)
192.168.255.138/32	10.30.0.4	OSPF via PA-FW DMZ-ISO loopback
10.40.0.0/24	10.30.0.4	OSPF via PA-FW DMZ-ISO (redistributed static)
0.0.0.0/0	Not set	No default route; originates default via OSPF

Cisco r2 vRouter (192.168.3.9)
Hostname: R2.home.com

Role: Edge router for ISO_LAN2.

Directly connected: 10.30.0.0/29, 192.168.3.0/24, 192.168.255.2/32 (Lo1)

OSPF adjacencies: R1 (10.30.0.1), R3 (10.30.0.3), PA-FW DMZ-ISO (10.30.0.4) — Area 0

Destination Subnet	Next Hop	Notes
10.30.0.0/29	Direct	P2P to R1, R3, PA-FW
192.168.3.0/24	Direct	ISO_LAN2
192.168.255.2/32	Direct	Lo1
192.168.1.0/24	10.30.0.1	OSPF via R1 (Prod_LAN)
192.168.100.0/24	10.30.0.1	OSPF via R1 (LAB_LAN1)
192.168.200.0/24	10.30.0.1	OSPF via R1 (LAB_LAN2)
192.168.2.0/24	10.30.0.3	OSPF via R3
192.168.255.1/32	10.30.0.1	OSPF via R1
192.168.255.3/32	10.30.0.3	OSPF via R3
10.20.0.0/24	10.30.0.4	OSPF via PA-FW DMZ-ISO
192.168.255.138/32	10.30.0.4	OSPF via PA-FW DMZ-ISO loopback
10.40.0.0/24	10.30.0.4	OSPF via PA-FW DMZ-ISO (redistributed static)
0.0.0.0/0	Not set	No default route configured

Cisco r3 vRouter (192.168.2.9)
Hostname: R3.home.com

Role: Edge router for DMZ.

Directly connected: 10.30.0.0/29, 192.168.2.0/24, 192.168.255.3/32 (Lo1)

OSPF adjacencies: R1 (10.30.0.1), R2 (10.30.0.2), PA-FW DMZ-ISO (10.30.0.4) — Area 0

Destination Subnet	Next Hop	Notes
10.30.0.0/29	Direct	P2P to R1, R2, PA-FW
192.168.2.0/24	Direct	DMZ
192.168.255.3/32	Direct	Lo1
192.168.1.0/24	10.30.0.1	OSPF via R1 (Prod_LAN)
192.168.100.0/24	10.30.0.1	OSPF via R1 (LAB_LAN1)
192.168.200.0/24	10.30.0.1	OSPF via R1 (LAB_LAN2)
192.168.3.0/24	10.30.0.2	OSPF via R2
192.168.255.1/32	10.30.0.1	OSPF via R1
192.168.255.2/32	10.30.0.2	OSPF via R2
10.20.0.0/24	10.30.0.4	OSPF via PA-FW DMZ-ISO
192.168.255.138/32	10.30.0.4	OSPF via PA-FW DMZ-ISO loopback
10.40.0.0/24	10.30.0.4	OSPF via PA-FW DMZ-ISO (redistributed static)
0.0.0.0/0	Not set	No default route configured

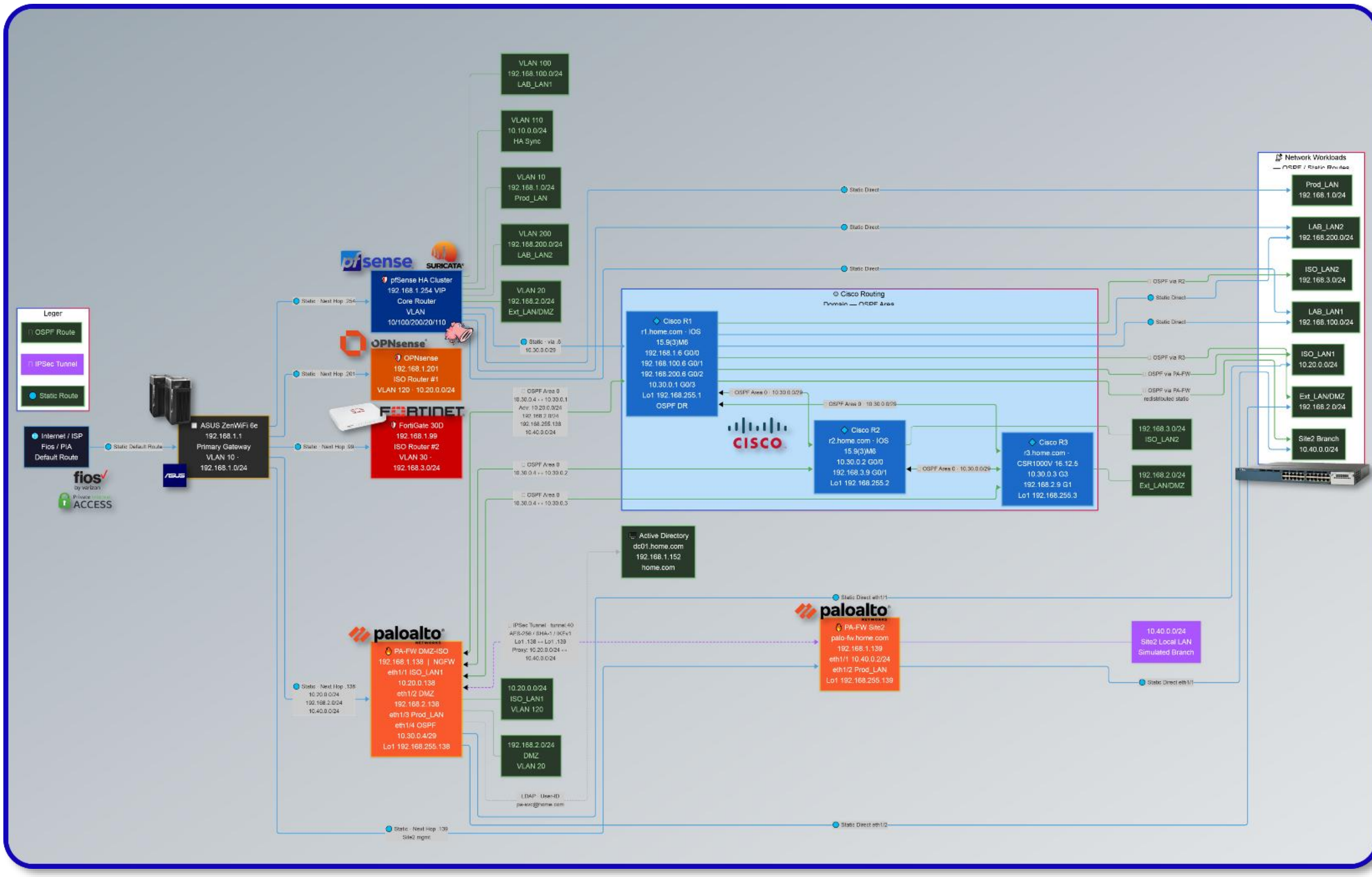
Tailscale Mesh VPN — Cross-Cloud Overlay

Tailscale provides the encrypted WireGuard overlay connecting on-premises nodes, cloud VMs, and remote endpoints. Each node receives a stable CGNAT address (100.64.0.0/10). pfSense nodes advertise on-premises subnets into the tailnet; cloud subnet routers advertise their respective VPC/VNet CIDRs.

Hostname	Platform	Local IP	Tailscale IP	Role
pfsense-a (fw.home.com)	pfSense FreeBSD	192.168.100.2	100.102.245.44	Subnet Router / Exit Node
pfsense-b (fw.home.com)	pfSense FreeBSD	192.168.100.3	100.74.247.76	Subnet Router / Exit Node
officepc.home.com	Windows 11	192.168.1.31	100.98.158.59	Endpoint
pve.home.com	Proxmox Host	192.168.1.178	100.79.235.93	Endpoint
aws-ec2-host1	Amazon Linux (AWS)	172.31.34.12	100.64.167.58	Subnet Router / Exit Node
aws-win2025-host2	Windows Server 2025 (AWS)	172.31.46.238	100.108.70.123	Endpoint
gcp-debian-host1	Debian 13.4 (GCP)	10.128.0.2	100.67.103.120	Subnet Router / Exit Node
az-ubuntu	Ubuntu 24.04 LTS (Azure)	10.130.0.7	100.96.110.40	Subnet Router / Exit Node
az-win2025-dc	Windows Server 2025 (Azure)	10.130.0.5	100.113.13.112	Endpoint

Subnet Advertisement

Node	Advertised Subnets	Exit Node
pfsense-a / pfsense-b	192.168.1.0/24, 192.168.100.0/24, 192.168.200.0/24	Yes
aws-ec2-host1	172.31.0.0/16	Yes
gcp-debian-host1	10.128.0.0/16	Yes
az-ubuntu	10.130.0.0/16	Yes



Domain Namespace Design

Domain	Purpose	Notes
home.com	Production / Lab services	Current state: Production, lab and AD environments. Future State: migration of domain to shadowitlab.com, AD will remain on home.com.
shadowitlab.com	Public DNS	Used to expose services externally through Cloudflare tunnels. Future migration to all internal DNS FQDNs.

Table A – Infrastructure Assets Summary Table

AssetID	Hostname	IP Address(s) (IPv4)	OS Platform, Distribution and Release	Purpose	FQDN	Resources (Cores/RAM /Storage)	Virtualization Host	Status	Ansible Enabled	OpenVAS Scanner	EDR Monitored (Wazuh)	Checkmk Monitored
KVM-VM: 100	UbuntuGenVM	192.168.1.126	Linux, Ubuntu Desktop: 25.04	General Lab		2 / 4 / 60	PVE	Online	✅	✅	✅	✅
KVM-VM: 101	Win11Pro	192.168.1.202	Windows, 11 Pro: 24H2 build 10.0.26100.4652	AD Domain PC1	win11pc.home.com	4 / 4 / 100	PVE	Online	❌	❌	✅	❌
KVM-VM: 102	KaliGenVM	192.168.1.100	Linux, Kali: 2025.2/Linux 6.12.25-amd64	General Lab	kali.home.com	2 / 4 / 60	PVE	Online	✅	✅	✅	❌
KVM-VM: 103	pfa	192.168.100.2(Lab LAN) 192.168.1.253 (Prod LAN) 192.168.200.2 (Lab2 LAN) 192.168.2.2 (EXT LAN) 192.168.100.1 (VIP-Lab LAN) 192.168.200.2 (VIP Lab2 LAN) 192.168.1.254 (VIP-Prod LAN) 192.168.2.1 (VIP-EXT LAN)	FreeBSD, FreeBSD: 15	Firewall for Lab LANs Segmentation	pfa.home.com, pf.home.com	2 / 4 / 32	PVE	Online	❌	✅	❌	❌
KVM-VM: 104	Ubuntu-pfLAN	192.168.2.6	Linux, Ubuntu Desktop: 25.04	Host for ExtLAN testing		2 / 4 / 100	PVE	Online	❌	✅	❌	❌
KVM-VM: 105	Win11-pfS	192.168.100.30	Windows, 11 Home: 24H2 build10.0.26100.4652	Host for Lab LAN1 testing		4 / 8 / 100	PVE	Offline	❌	❌	✅	❌
KVM-VM: 106	homeassistant	192.168.1.243	Home Assistant OS, Home Assistant OS: 16	Home Assistant OS		2 / 2 / 32	PVE	Offline	❌	✅	❌	❌
KVM-VM: 107	dc01	192.168.1.52	Windows, Sever 2022: 10.0.20348.3932		dc01.home.com	2 / 8 / 100	PVE	Online	❌	❌	✅	✅
KVM-VM: 108	win11pro2	192.168.1.182	Windows, 11 Pro WS: 24H2 build 10.0.26100.4652	AD Domain PC2	win11pc2.home.com	2 / 8 / 100	PVE	Online	❌	✅	✅	✅
KVM-VM: 109	pfb	192.168.100.3(Lab1 LAN) 192.168.1.249 (Prod LAN) 192.168.200.3 (Lab2 LAN) 192.168.2.3 (EXT LAN)	FreeBSD, FreeBSD: 15	HA Firewall for Lab LANs Segmentation	pfb.home.com, pf.home.com	2 / 4 / 32	PVE	Online	❌	✅	❌	❌
KVM-VM: 110	splunk	192.168.1.109	Linux, Ubuntu Live Server: 24.1	log aggregation and normalization	splunk.home.com	4 / 16 / 800	PVE	Online	✅	✅	❌	✅
KVM-VM: 111	dockervm2	192.168.1.166	Linux, Debian: 13	Docker host		2 / 8 / 100	PVE	Online	✅	✅	❌	✅
KVM-VM: 112	redhat	192.168.2.20	Linux, Red Hat: 10			2 / 8 / 32	PVE	Offline	❌	❌	❌	❌
KVM-VM: 113	fedora	192.168.100.5	Linux, Fedora: 43			2 / 8 / 100	PVE	Online	❌	❌	❌	✅
KVM-VM: 114	caine	192.168.1.33	Linux, Ubuntu: 24.04	Digital Forensics Test host		2 / 4 / 100	PVE	Offline	❌	❌	❌	❌
KVM-VM: 115	opnsense	192.168.1.201(PROD_LAN) 10.20.0.254 (ISO_LAN)	FreeBSD, FreeBSD: 14.3-RELEASE-p2	Firewall for ISO_LAN segmentation		2 / 2 / 100	PVE	Online	❌	❌	❌	❌
LXC-CT: 200	apache-ubuntu	192.168.1.108	Linux, Ubuntu: 24.1	Apache Webserver	web.home.com	1 / 1 / 16	PVE	Online	✅	✅	✅	✅
LXC-CT: 201	Plex-Ubuntu	192.168.1.136	Linux, Ubuntu: 22.04	Plex Server	plex.home.com	4 / 2 / 32	PVE	Online	✅	✅	✅	❌
LXC-CT: 202	Jellyfin-Ubuntu	192.168.200.244	Linux, Ubuntu: 22.04	Media Server	jellyfin.home.com	2 / 2 / 32	PVE	Online	✅	✅	❌	❌
LXC-CT: 203	Pi-hole-Ubuntu	192.168.1.250	Linux, Ubuntu: 22.04	ad blocker and DNS forwarder	pihole.home.com	1 / 0.5 / 32	PVE	Online	✅	✅	❌	✅
LXC-CT: 204	traefik	192.168.1.247	Linux, Debian: 12	Reverse Proxy	traefik.home.com	2 / 0.5 / 24	PVE	Online	✅	✅	❌	✅
LXC-CT: 205	Bind9-new	192.168.1.251	Linux, Ubuntu: 25.04	Local DNS	bind.home.com	2 / 0.5 / 8	PVE	Online	✅	✅	❌	✅
LXC-CT: 206	unbound	192.168.1.252	Linux, Ubuntu: 22.04	DNS resolver	unbound.home.com	2 / 0.5 / 32	PVE	Online	❌	✅	❌	✅
LXC-CT: 208	grafana-debian	192.168.1.246	Linux, Debian: 12	Analytics		2 / 1 / 6	PVE	Online	✅	✅	❌	❌
LXC-CT: 209	uptime-kuma-debian	192.168.1.181	Linux, Debian: 12	Application Health Status	uptimek.home.com	2 / 0.75 / 4	PVE	Online	✅	✅	❌	✅
LXC-CT: 210	kibana-debian	192.168.2.5	Linux, Debian: 12	Dashboard		2 / 0.75 / 8	PVE	Offline	❌	✅	❌	❌
LXC-CT: 211	ubuntu-pfS2	192.168.200.5	Linux, Ubuntu: 24.1	Host for Lab LAN2 testing		1 / 0.5 / 8	PVE	Offline	❌	❌	❌	❌
LXC-CT: 212	ubuntu-pfS2	192.168.100.4	Linux, Ubuntu: 24.1	Host for Lab LAN1 testing		1 / 0.5 / 32	PVE	Offline	❌	❌	❌	❌
LXC-CT: 213	vaultwarden	192.168.1.4	Linux, Debian: 12	password vault	vault.home.com	4 / 6 / 6	PVE	Online	✅	❌	❌	✅
LXC-CT: 214	Wazuh	192.168.1.219	Linux, Debian: 12	Security/EDP	wazuh.home.com	4 / 4 / 18	PVE	Online	✅	✅	❌	✅
LXC-CT: 215	stepca	192.168.100.51	Linux, Debian: 12	Online Signing CA	stepca.home.com	2 / 0.5 / 32	PVE	Online	✅	✅	❌	✅
LXC-CT: 219	overseerr	192.168.100.15	Linux, Debian: 12		ovrsr.home.com	2 / 2 / 8	PVE	Online	✅	✅	❌	❌
LXC-CT: 220	ansible	192.168.1.25	Linux, Debian: 12	Automation/SSH key deployment	ans.home.com	1 / 0.5 / 8	PVE	Online	❌	❌	❌	✅
LXC-CT: 221	centos	192.168.1.93	Linux, CentOS: Stream 9			2 / 0.5 / 24	PVE	Online	✅	✅	❌	❌
LXC-CT: 222	kms-iso	10.20.0.1	Linux, Ubuntu: 25.04	Host for ISOLAN testing		1 / 0.5 / 8	PVE	Online	✅	✅	❌	❌
LXC-CT: 223	crowdsec	192.168.1.33	Linux, Debian: 24.1	Threat Intelligence Engine		2 / 0.5 / 8	PVE	Online	✅	✅	❌	✅
LXC-CT: 250	rootca	192.168.100.50	Linux, Debian: 12	Certificate Authority	rootca.home.com	2 / 0.5 / 32	PVE	Offline	❌	❌	❌	❌
Physical: 1	pve	192.168.1.178 (NIC1) 192.168.100.10 (WIFI) 192.168.3.178 (LAG -ISO link), VLAN3	Linux, Debian: 12	Primary Lab Node				Online	❌	✅	❌	❌
Physical: 3	Logans-MacBook-Air	192.168.1.3	macOS, Sonoma: 14.7.8	Mac-based testing, forensics, scanning etc.				Online	❌	✅	✅	❌
Physical: 6	OfficePC2023	192.168.1.31 (eth0-2.5G) 192.168.1.51 (WIFI6) 192.168.3.31 (eth1-1G), VLAN3	Windows, 11 Pro: 24H2 build 10.0.26100.4652	Main Prod PC	officepc.home.com			Online	❌	✅	❌	✅
VMware VM: 2	pbs	192.168.1.239	Linux, Debian: 12	Application Backup Server	pbs.home.com		OfficePC	Offline	❌	❌	❌	❌
Vmware VM: 4	pbs	192.168.1.243	Linux, Debian: 13	Application Backup Server	pbs.home.com		OfficePC	Online	❌	❌	❌	❌
Vmware VM: 5	pms	192.168.1.x	Linux, Debian: 12	Email Security Gateway	pms.home.com		OfficePC	Offline	❌	❌	❌	❌
Vmware VM: 7	remnux	192.168.1.121	Linux, Ubuntu: 20.04 LTS	Malware analysis			OfficePC	Online	❌	❌	❌	❌

[Return to Proxmox Virtual Machine & Container Summary]

Table B – Application and Services Summary Table

Application and Version	Service Role	FQDN, Port Detailed	Deployed On (Asset ID)	Status	Log Aggregated (Splunk)	TLS Cert	OAuth2/ OIDC	Docker	Reverse Proxy
Proxmox pve-manager, Version: 8.4.5	Virtualization Platform	pve.home.com, Ports:8006(Internal):(External)	1	Running	✗	✓	✓	✗	✗
prometheus-pve exporter, Version:	exports data from pve to Prometheus		100	Running	✗	✗	✗	✓	✗
Checkmate, Version: 20.19.4	Checkmate backend server	chk-svr.home.com, Ports:52345(Internal):52345(External)	100	Config	✗	✓	✗	✓	✓
Checkmate, Version: 20.19.4	Checkmate client server	checkmate.home.com, Ports:80(Internal):7000(External)	100	Config	✗	✓	✗	✓	✓
Docker, Version: 28.3.3	Container Image Management		100	Running	✗	✗	✗	✗	✗
mongoDB, Version:	Checkmate DB		100	Config	✗	✗	✗	✓	✗
Nebula-Sync, Version: 0.11.1	Sync engine for Pihole		100	Running	✗	✗	✗	✓	✗
Wazuh Agent, Version: 4.12.0	Forwards data to Wazuh manager		100	Running	✗	✗	✗	✗	✗
pi-hole, Version: 2025.08.0	secondary instance - DNS ad blocker	piholebk.home.com, Ports:53, 80, 443(Internal):53, 80, 443(External)	100	Running	✗	✓	✗	✓	✓
PortainerCE, Version: 2.33.1 LTS	Docker Orchestration	portainer.home.com, Ports:8000, 9443(Internal):8000, 9443(External)	100	Running	✗	✓	✓	✓	✓
Prometheus , Version: 3.5	Scraper for Grafana monitoring	prom.home.com, Ports:9090(Internal):9090(External)	100	Running	✗	✗	✗	✓	✗
Pulse, Version: 4.10.0	Proxmox Monitor	pulse.home.com, Ports:7655(Internal):7655(External)	100	Running	✗	✓	✗	✓	✓
Tautulli, Version: 2.15.2	Media Server Add-on	tautulli.home.com, Ports:8181(Internal):8182(External)	100	Stopped	✗	✗	✗	✓	✓
Wazuh Agent, Version: 4.12.0	Forwards data to Wazuh manager		101	Running	✗	✗	✗	✗	✗
Wazuh Agent, Version: 4.12.0	Forwards data to Wazuh manager		102	Running	✗	✗	✗	✗	✗
CrowdSec firewall bouncer, Version: 1.7.0	Threat rule management		103	Running	✗	✗	✗	✗	✗
Node Exporter, Version: 0.18.1_5	exporter for Prometheus		103	Running	✗	✗	✗	✗	✗
OpenVPN (PIA), Version:	VPN		103	Running	✗	✗	✗	✗	✗
pfBlockerNG, Version: 3.2.8	Firewall Rules engine/DNS blocker		103	Running	✗	✗	✗	✗	✗
Service_Watchdog, Version: 1.8.7_4	pfSense service monitor		103	Running	✗	✗	✗	✗	✗
Snort, Version: 4.1.6_26	IDS/IPS		103	Running	✓	✗	✗	✗	✗
Splunk Forwarder, Version: 10.0.0	forwards logs to Splunk Indexer		103	Running	✗	✗	✗	✗	✗
Suricata, Version: 7.0.8_3	IDS/IPS		103	Running	✓	✗	✗	✗	✗
Tailscale, Version: 0.1.8	Mesh VPN		103	Running	✗	✗	✗	✗	✗
pfSense Virtual Firewall, Version: 2.8.1	Network Stateful Firewall and Router	fwa.home.com, fw.home.com, Ports:(Internal):(External)	103	Running	✓	✓	✗	✗	✗
Wazuh Agent, Version: 4.12.0	Forwards data to Wazuh manager		105	Running	✗	✗	✗	✗	✗
Docker, Version: 28.3.0	Container Image Management		106	Running	✗	✗	✗	✗	✗
Home Assistant Core, Version: 2025.7.4	Smart Home Dashboard		106	Running	✗	✗	✗	✓	✗
File and Storage Services, Version:		dc01.home.com, Ports:(Internal):(External)	107	Running	✗	✗	✗	✗	✗
Microsoft DNS, Version:	DNS for MS Domain	dc01.home.com, Ports:(Internal):(External)	107	Running	✗	✗	✗	✗	✗
Active Directory Domain Services, Version: Windows2016Domain/Forest	Identity and access management		107	Running	✓	✗	✗	✗	✗
Active Directory LDAP Services, Version:		dc01.home.com, Ports:(Internal):(External)	107	Running	✗	✗	✗	✗	✗

Table B – Continued.

Application and Version	Service Role	FQDN, Port Detailed	Deployed On (Asset ID)	Status	Log Aggregated (Splunk)	TLS Cert	OAuth2/ OIDC	Docker	Reverse Proxy
Splunk Forwarder, Version: 10.0.0	forwards logs to Splunk Indexer		107	Running	✗	✗	✗	✗	✗
Wazuh Agent, Version: 4.12.0	Forwards data to Wazuh manager		107	Running	✗	✗	✗	✗	✗
Windows Server Update Services, Version:		dc01.home.com, Ports:(Internal):(External)	107	Config	✗	✗	✗	✗	✗
OpenVPN (PIA), Version:	VPN		109	Running	✗	✗	✗	✗	✗
pfBlockerNG, Version: 3.2.8	Firewall Rules engine/DNS blocker		109	Running	✗	✗	✗	✗	✗
pfSense Virtual Firewall, Version: 2.8.1	Network Stateful Firewall and Router	fwb.home.com, fw.home.com, Ports:(Internal):(External)	109	Running	✓	✓	✗	✗	✗
Suricata, Version: 7.0.8_3	IDS/IPS		109	Running	✓	✗	✗	✗	✗
Tailscale, Version: 0.1.8	Mesh VPN		109	Running	✗	✗	✗	✗	✗
Splunk, Version: 10.0.0	SIEM	splunk.home.com, Ports:8000(Internal):(External)	110	Running	✗	✓	✗	✗	✓
Authentik Server, Version: 2025.6.4	Identity and access management	authentik.home.com, Ports:9000, 9443(Internal):9000, 9443(External)	111	Running	✗	✓	✗	✓	✓
Authentik Worker, Version: 2025.6.4	Identity and access management		111	Running	✗	✗	✗	✓	✗
Docker, Version: 28.3.3	Container Image Management		111	Running	✗	✗	✗	✗	✗
OpenVAS, Version: 25.3.0	The Greenbone Vulnerability Manager daemon		111	Running	✗	✗	✗	✓	✗
OpenVAS, Version: 25.3.0	A log-view container that tails the OpenVAS log		111	Running	✗	✗	✗	✓	✗
OpenVAS, Version: 25.3.0	The OpenVAS daemon		111	Running	✗	✗	✗	✓	✗
OpenVAS, Version: 25.3.0	The OSP (Open Scanner Protocol) wrapper for OpenVAS		111	Running	✗	✗	✗	✓	✗
OpenVAS, Version: 25.3.0	Greenbone Security Assistant (web UI)	vas.home.com, Ports:80(Internal):9392(External)	111	Running	✗	✓	✗	✓	✓
portainer agent, Version: 2.33.1	Connects to Portainer for remote management		111	Running	✗	✗	✗	✓	✗
PostgreSQL, Version: 16.1	database		111	Running	✗	✗	✗	✓	✗
PostgreSQL, Version:	openVAS DB		111	Running	✗	✗	✗	✓	✗
Redis, Version:	fast cache database		111	Running	✗	✗	✗	✓	✗
Redis, Version:	openVAS Redis		111	Running	✗	✗	✗	✓	✗
OPNsense, Version: 25.7.3	Firewall Rules engine/DNS blocker	opn.home.com, Ports:80, 443(Internal):(External)	115	Running	✓	✓	✗	✗	✗
Proxmox BS, Version: 3.4.3	Application Backup Server	pbs.home.com, Ports:8007(Internal):(External)	2	Running	✗	✗	✗	✗	✗
Wazuh Agent, Version: 4.12.0	Forwards data to Wazuh manager		200	Running	✗	✗	✗	✗	✗
Apache, Version: 2.4.62	internal web server/Dashboard	web.home.com, Ports:443(Internal):(External)	200	Running	✗	✓	✗	✗	✗
Wazuh Agent, Version: 4.12.0	Forwards data to Wazuh manager		201	Running	✗	✗	✗	✗	✗
Plex, Version: 1.41.9.9961	Media Server	plex.home.com, Ports:32400(Internal):(External)	201	Running	✗	✓	✗	✗	✓
Jellyfin, Version: 10.10.7	Media Server	jellyfin.home.com, Ports:8096(Internal):(External)	202	Running	✗	✗	✗	✗	✓
Docker, Version: 28.3.3	Container Image Management		203	Running	✗	✗	✗	✗	✗

Table B – Continued.

Application and Version	Service Role	FQDN, Port Detailed	Deployed On (Asset ID)	Status	Log Aggregated (Splunk)	TLS Cert	OAuth2/ OIDC	Docker	Reverse Proxy
pi-hole exporter, Version:	exports data from pihole to Prometheus		203	Running	❌	❌	❌	✅	❌
portainer agent, Version: 2.33.1	Connects to Portainer for remote management		203	Running	❌	❌	❌	✅	❌
pi-hole, Version: 2025.08.0	primary instance - DNS ad blocker	pihole.home.com, Ports:53, 80, 443(Internal):53, 80, 443(External)	203	Running	❌	✅	❌	✅	✅
Docker, Version: 28.3.3	Container Image Management		204	Running	❌	❌	❌	❌	❌
portainer agent, Version: 2.33.1	Connects to Portainer for remote management		204	Running	❌	❌	❌	✅	❌
Traefik, Version: 3.5.0	Reverse Proxy / secure web access	trfk.home.com, Ports:8080(Internal):(External)	204	Running	❌	✅	❌	✅	❌
Bind9, Version: 9.20.4	Authoritative DNS for local lookups	bind.home.com, Ports:(Internal):(External)	205	Running	❌	❌	❌	❌	❌
unbound, Version: 1.13.1	Recursive, cashing DNS resolver for external queries	unbound.home.com, Ports:53(Internal):(External)	206	Running	❌	❌	❌	❌	❌
dashy, Version:	dashboards		207	Stopped	❌	❌	❌	❌	❌
Wazuh Agent, Version: 4.12.0	Forwards data to Wazuh manager		208	Running	❌	❌	❌	❌	❌
grafana, Version: 12.1.1	monitoring dashboard for pve, pihole	grafana.home.com, Ports:(Internal):(External)	208	Running	❌	✅	❌	❌	✅
Uptime Kuma, Version: 1.23.16	service monitor	uptimek.home.com, Ports:3001(Internal):(External)	209	Running	❌	❌	❌	❌	❌
kibana, Version:	dashboards		210	Stopped	❌	❌	❌	❌	❌
Vaultwarden, Version: 2025.5.0	secrets management	vault.home.com, Ports:8000(Internal):(External)	213	Running	❌	✅	❌	❌	✅
Splunk Forwarder, Version: 10.0.0	forwards logs to Splunk Indexer		214	Running	❌	❌	❌	❌	❌
Wazuh, Version: 4.12.0	Wazuh manager	wazuh.home.com, Ports:1514, 1515(Internal):(External)	214	Running	✅	❌	❌	❌	❌
StepCA, Version: 0.26.1	Intermediate Certificate Authority	stepca.home.com, Ports:(Internal):(External)	215	Running	❌	❌	❌	❌	❌
overseerr, Version:	Media Search	ovrsr.home.com, Ports:5055(Internal):(External)	219	Running	❌	❌	❌	❌	❌
Git, Version: 2.39.5	git repository for ansible version control		220	Running	❌	❌	❌	❌	❌
MariaDB, Version: 15.1 Distrib 10.11.11-MariaDB	database		220	Running	❌	❌	❌	❌	❌
Semaphore, Version:	GUI for IoC solutions		220	Running	❌	❌	❌	❌	❌
Ansible, Version: 2.10.01	IaC Configuration	ans.home.com, Ports:12320, 12321(Internal):(External)	220	Running	❌	❌	❌	❌	❌
Terraform, Version: 1.12.2	IoC Platform		221	Running	❌	❌	❌	❌	❌
Apache, Version: 2.4.62	ISO Web Server		222	Running	❌	❌	❌	❌	❌
CrowdSec Engine, Version: 1.7.0	Local API engine for remediation components and loggers		223	Running	❌	❌	❌	❌	❌
CrowdSec firewall bouncer, Version: 1.7.0	Threat rule management		223	Running	❌	❌	❌	❌	❌
CrowdSec Logger, Version: 1.7.0	Local Logger for events		223	Running	❌	❌	❌	❌	❌
RootCA, Version:	certificate authority	rootca.home.com, Ports:(Internal):(External)	250	Stopped	❌	❌	❌	❌	❌

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